

AgentCore — Complete Product Requirement Document (PRD)

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PART 1: EXECUTIVE PRODUCT VISION

1.1 Product Vision Statement

AgentCore is the universal agentic AI platform that enables any customer-facing business to deploy voice, chat, WhatsApp, and email AI agents — connected to their existing tools and memory — in under 30 minutes, without writing a line of code.

1.2 Mission

Every business interaction today is a point of failure: a hold-queue erodes trust, a missed lead costs revenue, a manual follow-up doesn't happen. Human agents are expensive, inconsistent, available only during business hours, and cognitively limited to one conversation at a time. The AI revolution has produced impressive demos but failed to deliver production-grade, context-aware agents that handle real Indian business workflows — messy multi-lingual inputs, Exotel phone trees, WhatsApp UPI payments, and the regulatory requirements of DPDP, TRAI, and HIPAA.

AgentCore exists to eliminate the gap between AI's potential and its deployment reality. We build the infrastructure layer — the "brain" — so that a travel agency in Bangalore, a hospital in Hyderabad, and a fintech startup in Mumbai can all deploy AI that genuinely represents their business, remembers their customers, and executes real transactions. We don't sell chatbots. We sell cognitive infrastructure for businesses.

1.3 The Two-Sided Market Problem

Demand Side — Businesses:

Indian businesses spend an estimated ₹2.4 lakh crore annually on customer-facing operations: customer support, sales, follow-up, booking, and collections. The workforce handling these interactions is:

- High-churn (average contact center attrition: 30–45% annually)
- Inconsistent (quality varies hour-by-hour, agent-to-agent)

- Limited (8–10 hour coverage, no nights or weekends without overtime)
- Expensive at scale (mid-market companies spend ₹25,000–₹40,000/agent/month including salary, benefits, training, and attrition replacement cost)

Yet these same businesses cannot simply "buy" an AI solution. They face:

1. **Integration debt:** Their data is locked in vertical CRMs (ItinerarySuite for travel, Studio Ninja for photography, custom ERPs), not accessible by generic AI tools.
2. **Language complexity:** 30–40% of their customers communicate in Hindi, Tamil, Telugu, Bengali, or Kannada — no Western AI product handles this reliably.
3. **Regulatory friction:** DPDP, TRAI DND, and industry-specific compliance (HIPAA for healthcare, PCI for fintech) create legal risk for unvetted AI deployments.
4. **Trust deficit:** Businesses don't trust black-box AI to represent them to customers without explainability and override capability.

Supply Side — AI Capabilities:

The AI toolchain has matured dramatically: GPT-4.1-mini delivers 90%+ accuracy on structured tasks at \$0.40/M tokens, Deepgram Nova-3 delivers 97%+ STT accuracy at \$0.0043/min, Sarvam-M105B handles 11 Indic languages for free. LiveKit enables WebRTC at sub-400ms latency on \$49/month infrastructure. The raw materials for production-grade AI agents are now commodity.

The Gap AgentCore Bridges:

Neither category of existing solutions bridges this gap. Generic AI platforms (OpenAI GPT, Google Gemini) provide raw capability but no business workflow scaffolding, no vertical integration, no multi-tenant architecture, no compliance layer. Pure contact center platforms (Genesys, Avaya) have deep workflow tooling but no native Indian language AI, prohibitive enterprise pricing (₹15–30 lakhs/year), and 6–12 month implementation timelines.

AgentCore uniquely occupies the intersection: genuine AI capability (hybrid voice pipeline, 4-layer memory, 7-agent orchestration) with real business workflow tooling (24 engines covering the full CRM lifecycle), deployed in a multi-tenant SaaS model that SMBs can self-serve and enterprises can white-label.

1.4 Strategic Bets

Bet 1 — India is the test lab, not an afterthought.

Every Western AI company builds for English first, adds Indian language support as a translation layer. We build Sarvam-first, Exotel-first, UPI-first, DPDP-first. The 500M+ Indian businesses that will digitize over the next decade will adopt AI that was designed for them. Being genuinely India-native is a 3–5 year moat that cannot be replicated by appending a Sarvam API call to an existing Western product.

Bet 2 — Memory is the differentiator, not generation.

Every AI company competes on which model generates the most fluent response. We compete on which platform remembers the most useful context. A customer who complained about a billing error 6 months ago and then calls about a refund should be handled with that history present. Our 4-layer memory architecture (Redis → episodic → pgvector semantic → procedural rules) is the only

system in our competitive set that delivers cross-session, cross-channel personalization at the data layer rather than the prompt engineering layer.

Bet 3 — The platform business beats the product business.

We will resist the temptation to build only vertical SaaS (ItinerarySuite, PhotoAdda). The platform-as-infrastructure model (Model A API, Model B White-Label) reaches a 10x larger addressable market. Every partner who white-labels AgentCore becomes a distribution channel. Every API customer becomes a case study. We invest in developer experience, webhook ecosystems, and partner portals as first-class product work.

Bet 4 — Compliance is a feature, not a tax.

Indian enterprises — especially BFSI and healthcare — are paralyzed by AI adoption because no vendor offers DPDP + TRAI + HIPAA + PCI compliance out of the box. Being the first Indian AI platform with a comprehensive compliance framework (SOC 2, HIPAA BAA, TRAI disclosure rules baked into the product) converts regulatory risk from a blocker to a sales accelerator. We will publish compliance documentation before we publish pricing.

Bet 5 — Agentic beats chatbot every time customers see the demo.

The market is filled with chatbots that answer FAQs. AgentCore's agents execute multi-step workflows: they look up a booking, apply a discount, send a WhatsApp confirmation, and create a refund case — in one conversation, across voice and chat. The "wow moment" in any demo converts when the prospect sees the agent actually *do something* rather than just answer a question. We build toward agentic execution as the core product value, not as a future roadmap item.

1.5 Success in 3 Years

Revenue: ₹25 crore ARR by April 2029. Breakdown: ₹12 crore from Model A API customers (120+ tenants at average ₹1 lakh/month), ₹8 crore from Model B White-Label partners (20 partners × 20 sub-tenants average × ₹2,000/month), ₹5 crore from Model C vertical SaaS products (ItinerarySuite, PhotoAdda, aayiq.ai embedded revenue).

Product: AgentCore processes 50 million AI-handled interactions per month. Average AI resolution rate across all tenants: ≥75% (human not required). Mean customer satisfaction (CSAT) score on AI-handled interactions: ≥4.1/5.

Platform: 10,000+ active tenants on the platform including sub-tenants. 50+ certified integration partners. OpenAPI ecosystem with 200+ community-built integrations. SOC 2 Type II certified. Listed on AWS Marketplace and listed as a Sarvam AI partner.

Market Position: Top 3 most-cited AI customer engagement platform in India (per Tracxn, G2, and Nasscom reports). Two major enterprise reference customers (5,000+ employees) publicly referenceable. At least one international deployment (UAE, Singapore, or UK).

PART 2: MARKET ANALYSIS

2.1 Total Addressable Market (TAM)

The global AI in customer service market was valued at approximately \$11.5 billion USD (₹96,000 crore) in 2024 and is projected to reach \$47.8 billion USD (₹4 lakh crore) by 2030 at a 26.1% CAGR.

India-specific TAM by vertical:

Industry	Businesses (India)	AI-Addressable Revenue	TAM (INR)
Travel & Tourism	1.2 lakh+	₹8,000/business/month	₹11,520 crore/year
Healthcare	80,000+ clinics/hospitals	₹6,000/business/month	₹5,760 crore/year
BFSI (Insurance + Lending)	15,000+ entities	₹40,000/business/month	₹7,200 crore/year
Real Estate	90,000+ firms	₹5,000/business/month	₹5,400 crore/year
Education & Coaching	2 lakh+ institutes	₹3,000/business/month	₹7,200 crore/year
Retail & eCommerce	5 lakh+ SMBs	₹2,000/business/month	₹12,000 crore/year
Logistics & Delivery	25,000+ firms	₹10,000/business/month	₹3,000 crore/year
Automotive	30,000+ dealerships	₹8,000/business/month	₹2,880 crore/year
Hospitality & Food	1 lakh+	₹3,500/business/month	₹4,200 crore/year
Photography & Events	3 lakh+ studios	₹1,500/business/month	₹5,400 crore/year
Total India TAM			₹64,560 crore/year (~\$7.7B USD)

Global TAM (AgentCore addressable geography): \$47.8B USD by 2030, with South Asia, Middle East, and Southeast Asia representing ₹3.2 lakh crore of accessible market.

2.2 Serviceable Addressable Market (SAM)

AgentCore's SAM is the subset of TAM served by SaaS AI platforms in the ₹5,000–₹1,00,000/month price band — i.e., businesses that have digitized sufficiently to integrate an API or use a web-based platform, but are not large enough to build AI in-house.

India SAM (Year 1–3 horizon):

- Travel: 15,000 agencies with digital booking systems → SAM ₹1,440 crore/year
- Healthcare: 8,000 mid-market clinics/hospital networks → SAM ₹576 crore/year
- BFSI: 3,000 insurance agencies, NBFCs, fintechs → SAM ₹1,440 crore/year
- Real Estate: 10,000 digitized agencies → SAM ₹600 crore/year
- Education: 25,000 coaching institutes with CRM → SAM ₹900 crore/year
- Other verticals: ~20,000 businesses → SAM ₹1,200 crore/year

Total India SAM: ₹6,156 crore/year (~\$740M USD)

Global SAM (Year 3 expansion):

English-speaking markets (UK, US, UAE, Singapore, Canada) add 3× multiplier on the India SAM.

Global SAM: ₹18,468 crore/year (~\$2.2B USD)

2.3 Serviceable Obtainable Market (SOM)

Year	Tenants	Avg ARR/Tenant	Total ARR	Market Share of India SAM
Year 1 (FY2027)	150	₹5,00,000	₹7.5 crore	0.12%
Year 2 (FY2028)	600	₹7,00,000	₹42 crore	0.68%
Year 3 (FY2029)	1,500	₹8,50,000	₹127.5 crore	2.07%

Year 1 SOM is conservative and deliberately so — it requires 150 paying tenants, achievable with just 3 enterprise deals at ₹1 lakh/month, 15 growth customers at ₹30,000/month, and 100 starter customers at ₹10,000/month. The ramp from Year 1 to Year 3 is driven primarily by the [aayiq.ai](#) PLG motion adding sub-₹10,000/month SMB customers at volume.

2.4 Market Timing: Why Now?

- 1. Indian language AI reached production-grade quality in 2025–2026.** Sarvam Saaras V3 and Sarvam-M105B deliver accuracy on par with English models for 11 Indic languages — a capability that simply did not exist at this cost point in 2023.
- 2. DPDP Act 2023 enforcement began in 2025.** Every business handling customer data in India now has legal obligations around consent, purpose limitation, and data erasure. This creates a buying trigger: compliance-ready AI platforms become mandatory, not optional.
- 3. Post-COVID, Indian consumers expect 24×7 digital interaction.** A 2025 Redseer study found 68% of Indian consumers expect same-day response to support queries, with 34% expecting response within 1 hour. Human-only operations cannot meet this expectation economically.
- 4. WhatsApp Business API matured in 2024–2025.** Meta's introduction of Flows v2, in-chat UPI payments, and the 360dialog/AiSensy BSP ecosystem made WhatsApp a genuine transactional channel — not just messaging. This creates the infrastructure for agentic commerce at massive scale.
- 5. LLM costs dropped 40× in 18 months.** GPT-4.1-mini at \$0.40/M tokens makes AI resolution economically superior to human handling for interactions above ~4 minutes. The unit economics that didn't work in 2023 work convincingly today.

2.5 Customer Segmentation

SMB (Starter Tier — ₹10,000/month):

- Revenue: ₹1–10 crore/year
- Team size: 2–25 employees
- Technology: Basic CRM or spreadsheets, WhatsApp Business app, no API integrations
- Decision maker: Owner/founder, makes purchase in <2 weeks
- Key buying criterion: "Will this pay for itself in 1 month?"
- Primary channels: [aayiq.ai](#) PLG, digital ads, YouTube tutorials

- Primary use case: Replace after-hours call handling and WhatsApp follow-ups

Mid-Market (Growth Tier — ₹30,000/month):

- Revenue: ₹10–100 crore/year
- Team size: 25–250 employees
- Technology: Vertical CRM, some API capability, dedicated IT person
- Decision maker: Operations manager or CXO, 4–8 week sales cycle
- Key buying criterion: "Will this reduce my headcount cost and improve SLA?"
- Primary channels: Partner referrals, industry events, LinkedIn outreach
- Primary use case: AI first-line across all channels + human escalation for complex cases

Enterprise (Scale/Enterprise Tier — ₹75,000–1,00,000/month):

- Revenue: ₹100 crore+/year
- Team size: 250+ employees
- Technology: Salesforce/SAP, custom integrations, InfoSec review required
- Decision maker: CTO + Head of CX, 12–24 week sales cycle
- Key buying criterion: "Is this SOC 2 certified? Can we run it in our VPC?"
- Primary channels: Direct sales, Systems Integrator partnerships, Model A API
- Primary use case: Full agentic orchestration, compliance, white-label or API integration

PART 3: USER PERSONAS

Persona 1 — Rahul Mehta, Travel Agency Owner (SMB)

Demographics: 38 years old, Bangalore. Owns a mid-sized travel agency specializing in domestic packages and corporate travel. Annual revenue: ₹3.2 crore. Team of 8.

Daily Workflow: Rahul spends his mornings handling WhatsApp inquiries (60–80 messages/day), afternoons on quote preparation and vendor negotiation, evenings following up on pending bookings. He uses ItinerarySuite for CRM but receives leads from multiple channels — Instagram, Justdial, WhatsApp, phone. He misses calls constantly when on client calls.

Key Frustrations:

- Misses 30–40% of inbound calls during business hours; missed calls never call back
- Quotes take 45 minutes to prepare manually; competitors respond faster
- Follow-up is inconsistent — "I know I should call Mr. Sharma back but I forget"
- Can't scale his team affordably; each new agent costs ₹25,000/month and takes 3 weeks to train

Current Solutions: ItinerarySuite CRM, WhatsApp Business app, Exotel basic IVR, Google Sheets for follow-ups.

What Rahul Needs from AgentCore:

- AI agent answers every call 24×7 in Hindi and English, qualifies the lead, and books a callback

- Auto-generates quotes from the conversation and sends via WhatsApp
- Proactive follow-up calls for pending proposals without Rahul remembering to trigger them

Jobs to be Done:

- "When I'm on a call, make sure no lead is lost"
- "Prepare and send my standard Manali package quote automatically"
- "Follow up with prospects who went quiet after 2 days"

Persona 2 — Priya Krishnamurthy, Hospital Administrator (Mid-Market)

Demographics: 45 years old, Chennai. Administrator of a 150-bed multi-specialty hospital. Reports to Medical Director. Manages 12 front-desk staff plus 3 call center agents.

Daily Workflow: Priya begins her day reviewing appointment no-show rates (typically 18–22%), then checks patient complaint logs, then reviews insurance pre-authorization status. Her team handles 300+ inbound calls daily across OPD scheduling, billing queries, and lab report requests. 40% of calls are simple, repetitive, and could be handled without a human.

Key Frustrations:

- No-show rate costs ₹8–12 lakh/month in unutilized appointment slots
- Patients call for lab results 3–5 times per report — each call takes 4 minutes
- Staff burnout from repetitive calls; high attrition (35% annually)
- Patient complaints about hold times (average: 7 minutes)
- HIPAA-equivalent obligations under DPDP mandate data handling she can't audit

Current Solutions: Hospital Information System (HIS), manual appointment scheduling, Exotel IVR, WhatsApp for appointment reminders.

What Priya Needs:

- AI agent that confirms/reschedules appointments via voice and WhatsApp automatically
- Lab result notification via WhatsApp with patient authentication
- HIPAA BAA from vendor; auditable data access logs

Jobs to be Done:

- "Reduce no-show rate by 40% through automated confirmation and reminder calls"
- "Handle lab result queries without staff involvement"
- "Prove to our legal team that patient data is secure and DPDP-compliant"

Persona 3 — Kiran Bhat, CTO at Fintech Startup (Enterprise)

Demographics: 34 years old, Pune. CTO of a Series B lending platform (₹200 crore ARR, 150-person team). Reports to CEO. Oversees tech team of 45 engineers.

Daily Workflow: Kiran manages three engineering squads — product, platform, and data. His team built a loan origination platform in-house but needs AI-driven customer communication for collections, KYC reminders, and EMI queries. His current contact center (30 agents, outsourced) costs ₹45 lakh/month and underperforms on CSAT (3.2/5).

Key Frustrations:

- Building conversational AI in-house takes 6–9 months and requires 3–4 ML engineers he can't hire
- Existing contact center BPO has no context on their loan lifecycle — each call starts from zero
- Collections calls are inconsistent and generate regulatory complaints
- Can't integrate a third-party tool without InfoSec review, VAPT, and legal sign-off

Current Solutions: Custom-built ticketing, Exotel telephony, outsourced BPO, homegrown WhatsApp notification system.

What Kiran Needs:

- REST API + WebSocket API to embed AgentCore's voice AI in his platform
- Memory that knows the loan lifecycle, payment history, and past complaints for each customer
- TRAI-compliant collections workflow with full audit trail
- VAPT report, SOC 2 certification, and data processing agreement

Jobs to be Done:

- "Give me an API so I can add AI voice calling to my product in 2 sprints"
- "Make sure the collections agent knows everything about this customer before it calls"
- "Help me pass our board's security review for AI vendor adoption"

Persona 4 — Meena Joshi, Photography Studio Owner (SMB)

Demographics: 32 years old, Mumbai. Owns a 6-year-old photography studio specializing in weddings and corporate events. Revenue: ₹85 lakh/year. Team: herself + 3 photographers + 1 part-time admin.

Daily Workflow: Meena's business is highly seasonal. During wedding season (Nov–Feb), she handles 15–25 inquiry calls per day. She misses most of them while on shoots. Each missed call is potentially a ₹1–5 lakh booking. Her admin works 10am–6pm only. Quotations require customization based on package, location, and date.

Key Frustrations:

- Misses 40–50% of calls during shoots
- Prospects call 3–4 competitors; whoever responds first wins the booking
- Quote preparation takes 30–45 minutes of admin time
- No systematic follow-up on sent quotes (admin manually tracks on spreadsheet)

Current Solutions: PhotoAdda CRM (ItinerarySuite sister product), WhatsApp Business, Instagram DM, basic call forwarding.

What Meena Needs:

- AI agent that picks up every call, understands the inquiry (wedding? corporate? date? location?), and sends a preliminary quote via WhatsApp instantly
- Automated follow-up sequence: call → quote → reminder at D+2 → closing call at D+5
- Integration with PhotoAdda CRM so all leads appear automatically

Jobs to be Done:

- "Never miss an inquiry again — even at 11pm during a wedding"
- "Send my standard wedding package quote automatically before I call back"
- "Follow up so I don't have to remember who I sent quotes to"

Persona 5 — Anil Kapoor, Contact Center Head at Insurance Company (Enterprise)

Demographics: 51 years old, Mumbai. Head of Customer Operations at a mid-size general insurance company (₹1,200 crore premium). Manages 350 agents across 3 centers. Reports to COO.

Daily Workflow: Anil manages call center KPIs: AHT (average handle time, currently 8.5 min), FCR (first call resolution, currently 64%), CSAT (3.6/5), and agent utilization (78%). His team handles policy queries, claims status, renewal reminders, and complaints. 60% of calls are Tier-1 (simple, repetitive); only 40% require human expertise.

Key Frustrations:

- ₹1.2 crore/month labor cost for Tier-1 calls that AI should handle
- Agent onboarding takes 6 weeks; high churn means perpetual training overhead
- Compliance risk: agents don't consistently follow regulatory scripts for disclosures
- No real-time coaching for agents on complex calls — supervisors are stretched

Current Solutions: Genesys CX Cloud (₹35 lakh/year), Salesforce Service Cloud, homegrown quality monitoring tool.

What Anil Needs:

- White-label AI agent that handles his full Tier-1 call volume under his brand (not "AgentCore")
- Compliance engine that mandates IRDAI disclosures and TRAI consent before every call
- Agent Copilot that suggests responses in real time for human agents on complex calls
- Full audit trail for regulatory review

Jobs to be Done:

- "Automate 60% of my call volume at 40% of current cost"
- "Ensure every call is compliant without relying on agent memory"
- "Give my supervisors real-time visibility into every conversation"

PART 4: PROBLEM STATEMENT

4.1 The 8 Universal Customer Service Failures

Failure 1 — The Missed Call

In India, 35–45% of inbound business calls go unanswered during business hours and nearly 100% go unanswered after hours. A missed call from a prospect costs ₹500–₹50,000 in lost revenue depending on the industry. A missed call from an existing customer costs an average of 1.8 CSAT points. Current workaround: IVR voicemail that 80% of callers hang up on before leaving a message. Zero callbacks because there is no system.

Failure 2 — The Knowledge Gap

The average new contact center agent takes 6–8 weeks to reach baseline competency, yet handles live customer calls from week 1. The knowledge gap between what the customer needs and what the agent knows results in 36% first-call-non-resolution (FCNR) industry-wide. Each FCNR costs ₹180 (additional handle time) and 0.7 CSAT points. Current workaround: knowledge bases that agents don't have time to search during calls.

Failure 3 — The Memory Failure

A customer who called last week about a complaint is treated as a stranger on the next call. Every interaction starts from zero: name, account number, issue re-explanation. Average "re-identification" time: 90 seconds per call. For a contact center handling 1,000 calls/day, that is 25 agent-hours/day wasted on re-identification — ₹35,000/day in labor cost.

Failure 4 — The Channel Silos

A customer who raised a complaint on WhatsApp and then calls is told "we have no record of your message." The average business uses 3.8 separate tools for customer communication (WhatsApp Business, email, phone, chat), none of which share context. This creates 22% "channel bounce" — customers who contact through a second channel because they received no response on the first.

Failure 5 — The Follow-Up Black Hole

70% of sales leads in India receive zero follow-up after initial contact (Salesforce India SMB Report, 2024). Agents intend to follow up but are pulled into handling inbound volume. Systematic outbound follow-up is a capability only mature BPOs can execute, at costs SMBs cannot afford. The average travel agency loses 40% of interested prospects to competitors who responded faster.

Failure 6 — The Compliance Landmine

TRAI DND regulations require businesses to check suppression lists before outbound calls. IRDAI mandates specific disclosures before insurance policy discussions. DPDP Act 2023 requires explicit consent for data processing and call recording. Non-compliance fines range from ₹10,000 to ₹25 crore depending on severity. Current workaround: manually trained agents who forget or skip disclosures under pressure.

Failure 7 — The Peak Load Failure

During sales seasons, promotional periods, or crisis events, inbound volume spikes 3–5× above baseline. Human agent capacity cannot flex this fast. A travel agency during Dussehra promotions may receive 200 calls per hour with 8 agents available — 95% of calls go unanswered. Current workaround: "Please call back" IVR messages that permanently damage brand trust.

Failure 8 — The Escalation Abyss

When customers escalate to supervisors or managers, average wait time is 12–18 minutes. 45% of escalations could have been resolved at the first tier with better knowledge access. When escalation does happen, supervisors receive no contextual briefing — the customer must re-explain the entire situation. Current workaround: manual note-taking by the first agent that the supervisor reads while the customer waits.

4.2 Why Existing Solutions Fail

Category: Generic AI Chatbots (Intercom Fin, Freshdesk Freddy)

Miss: No voice. No Indic language support. No transactional workflow execution (can't book, can't generate a quote, can't process a refund). Trained on English FAQs, fail on complex Indian business workflows. Cannot be deployed in a contact center telephony stack (Exotel/Twilio).

Category: Contact Center Platforms (Genesys, NICE, Avaya)

Miss: ₹15–30 lakh/year minimum, 6–12 month implementation, zero support for Indian languages, no WhatsApp native integration, no LLM-native architecture. Architected for Fortune 500 enterprise IT, completely inaccessible to SMBs and mid-market companies. No agentic execution — agents only route and record, they don't execute.

Category: Indian CRM/Helpdesk (Freshdesk, Zoho Desk)

Miss: Strong workflow tooling but no AI voice agent, no native memory architecture, no proactive outbound AI, no multi-agent orchestration. WhatsApp integration is basic notification-only, not conversational commerce.

Category: WhatsApp Marketing Tools (AiSensy, Interakt, WATI)

Miss: One-way or template-only messaging. No voice. No LLM-based conversational resolution. No CRM lifecycle management. Treated as a broadcast channel, not an agentic customer service channel.

Category: AI Voice Platforms (ElevenLabs, Voiceflow, Bland AI)

Miss: English-only or weak Indic support. No multi-channel (chat, WhatsApp, email). No built-in CRM lifecycle management. No compliance layer for Indian regulations. No multi-tenant architecture for white-labeling.

4.3 The Gap AgentCore Fills

AgentCore occupies a completely unique market position: **agentic AI execution** (not just response generation) across **all four channels** (voice, chat, WhatsApp, email), with **genuine Indic language capability** (Sarvam-native), **CRM lifecycle integration** (all 24 engines), **compliance by default** (DPDP, TRAI, HIPAA), and **three deployment modes** (API, white-label, embedded SaaS). No single competitor covers more than three of these five dimensions simultaneously.

The competitive gap is widest in the mid-market segment (₹10–100 crore revenue businesses) where Genesys is too expensive, Intercom is too generic, AiSensy is too limited, and custom development is too slow.

PART 5: PRODUCT GOALS & NON-GOALS

5.1 Product Goals

1. **Reduce first-response time** from average 4.2 hours to under 30 seconds for all inbound inquiries across voice, chat, and WhatsApp, within 6 months of tenant activation.
2. **Achieve 75%+ AI resolution rate** (conversations resolved without human agent involvement) across all tenants within 12 months of launch, measured as the ratio of AI-closed conversations to total conversations.
3. **Reduce tenant cost-per-interaction** by 60% compared to human-only operation within 18 months, measured by comparing per-interaction labor cost before/after AgentCore deployment using tenant-reported data.
4. **Deliver <800ms end-to-end voice response latency** (STT + LLM + TTS) for 95th percentile of conversations in the cascaded pipeline, measured continuously via production telemetry.
5. **Achieve 99.9%+ uptime** for all Growth and Scale tier tenants within 9 months, measured as monthly availability excluding planned maintenance windows.

5.2 Non-Goals

1. **AgentCore will NOT build its own LLM or STT model.** We are a platform that orchestrates best-in-class AI models (GPT-4.1-mini, Sarvam-M, Deepgram). Building proprietary models requires \$50M+ in training infrastructure and 20+ ML researchers — not our business. We invest in the orchestration layer, not the model layer.
2. **AgentCore will NOT operate as a BPO or managed service.** We provide the AI platform and tooling; customers configure and operate their own agents. We offer professional services for onboarding and customization but do not take on SLA responsibility for the content or outcome of customer conversations.
3. **AgentCore will NOT build a native mobile app for end customers.** The Mobile SDK enables our tenants to embed AgentCore into their own apps. We do not build a consumer-facing AgentCore mobile app. This is a B2B platform.
4. **AgentCore will NOT support social media monitoring or brand listening.** We handle conversational channels (voice, chat, WhatsApp, email, Instagram DM, Facebook Messenger) but we are not a social media management tool. Twitter/X monitoring, sentiment analysis across social channels, and influencer tracking are out of scope.
5. **AgentCore will NOT offer industry-specific clinical decision support, legal advice generation, or financial planning advice.** Our AI agents assist with operational tasks (booking, quoting, support, follow-up). For regulated advice-giving (prescriptions, investment recommendations, legal opinions), we provide escalation to human experts, not AI-generated advice.

5.3 Success Metrics — AARRR Framework

Acquisition

- Target: 150 paying tenants by Month 12
- Measurement: New tenant activations per month (tracked in billing system)
- Timeline: Monthly review; quarterly board reporting

Activation

- Target: 80% of new tenants complete their first AI conversation within 48 hours of signup
- Measurement: Time-to-first-conversation event in product analytics (Mixpanel)
- Timeline: Measured per cohort; weekly engineering review

Retention

- Target: Month-12 tenant retention $\geq 85\%$; monthly churn $< 1.5\%$
- Measurement: Active tenants with ≥ 100 conversations in the billing period / tenants active 30 days prior
- Timeline: Monthly churn report

Revenue

- Target: ₹7.5 crore ARR by Month 12; ₹25 crore ARR by Month 36
- Measurement: MRR tracked in billing system; $ARR = MRR \times 12$
- Timeline: Weekly finance review

Referral

- Target: NPS ≥ 50 by Month 12; 25% of new tenants sourced from referral by Month 18
- Measurement: In-product NPS survey at Day 30 and Day 90; referral attribution in CRM
- Timeline: Quarterly NPS survey; monthly attribution review

PART 6: COMPLETE FEATURE REQUIREMENTS

Group A — Communication Engines

Engine 1: VoiceAgent Engine

Problem Solved: Businesses miss inbound calls and cannot scale human agent capacity to match call volume. AI voice agents must handle calls 24x7 in Hindi and English with $< 800\text{ms}$ response latency.

User Stories:

- As Rahul (travel agency owner), I want an AI agent to answer every call when I'm busy, so that no lead is ever lost.
- As Priya (hospital administrator), I want patients to confirm or reschedule appointments via voice, so that my staff can focus on in-person patients.

- As Kiran (fintech CTO), I want to trigger outbound AI voice calls via API with a pre-loaded customer context, so that my collections team can focus on complex disputes.
- As Anil (insurance contact center head), I want the AI to handle Tier-1 policy queries and escalate only complex cases, so that my agents handle only high-value work.
- As any tenant admin, I want to configure the AI voice persona (name, accent, language), so that the agent sounds like our brand.

P0 Requirements:

- Inbound call handling via Exotel (India) and Twilio (global) with <2s ring-to-answer time
- Hybrid pipeline: cascaded STT+LLM+TTS (600–1000ms) and S2S via OpenAI Realtime (200–400ms), with automatic mid-call switching based on latency
- English STT via Deepgram Nova-3; Indic STT via Sarvam Saaras V3 (11 languages)
- Language auto-detection within first 3 seconds of speech
- Barge-in detection: agent stops speaking when customer starts, within 200ms
- Pre-call TRAI consent disclosure (configurable per tenant)
- Call recording with customer consent; stored in S3 with tenant-scoped access
- Post-call transcript and summary generated automatically
- Human escalation trigger: "transfer to agent" keyword OR confidence < 0.6 for 3 turns
- 4 escalation modes: warm transfer, cold transfer, callback queue, voicemail

P1 Requirements:

- Voice cloning: tenant uploads 30 seconds of audio → ElevenLabs clone → stored as `voice_id` per tenant
- Custom IVR menu before AI agent (press 1 for English, 2 for Hindi, 3 for Sales)
- Call whispering: supervisor can speak to agent without customer hearing
- Call barging: supervisor joins a live call in 3-party mode
- DTMF (key press) input support for IVR menus

P2 Requirements:

- Acoustic echo cancellation for noisy call center environments
- Background noise suppression (office noise, street noise)
- Call sentiment analysis with real-time supervisor alert on negative sentiment

Acceptance Criteria :

- Given a customer calls a tenant's number, when the call is routed to AgentCore, then the AI agent answers within 2 seconds of the first ring.
- Given an active call in cascaded mode, when pipeline latency exceeds 1,200ms for two consecutive turns, then the system automatically switches to S2S mode within the next turn.
- Given a customer speaks in Hindi after an English greeting, when language detection confidence exceeds 0.85, then the AI switches to Sarvam Saaras V3 + Sarvam Bulbul v3 for all subsequent turns.

- Given a customer says "connect me to a human" or expresses frustration three times, when the escalation trigger fires, then the call transfers to an available agent within 30 seconds with a warm briefing.
- Given a call is recorded, when the customer presses 9 to opt out of recording, then recording stops within 1 second and a confirmation message plays.

Dependencies: 4-Layer Memory Engine (session context), Compliance Engine (TRAI consent), Tenant Management (voice persona config), Case Management (escalation case creation).

Engine 2: ChatAgent Engine

Problem Solved: Website visitors and app users need immediate conversational assistance. The chat agent must resolve queries, escalate to humans, and execute transactions — not just answer FAQs.

User Stories:

- As a hotel guest visiting the hotel website, I want to check room availability and book via chat, so that I don't need to call or navigate to an external booking site.
- As Priya (hospital admin), I want patients to check appointment status and receive lab result links via chat widget, so that 40% of inbound calls are deflected.
- As a tenant developer, I want to embed the chat widget with a single `<script>` tag, so that I can go live in 15 minutes without backend work.
- As a customer service agent, I want to see real-time AI-suggested replies and relevant KB articles, so that I can respond faster and more accurately.
- As an admin, I want to set proactive chat triggers based on page URL and dwell time, so that the widget engages visitors at the right moment.

PO Requirements:

```
- CDN embed: `<script src="https://cdn.agentcore.ai/v1/loader.js" data-tenant="TENANT_ID"></`
```

- JavaScript `AgentCore.init({tenantId, theme: {primary, secondary}, position: 'bottom-right'|'bottom-left', greeting, channels: ['chat', 'voice', 'whatsapp']})` API
- Shadow DOM isolation to prevent tenant CSS from breaking widget styles
- Widget states: minimized bubble (unread count badge), expanded chat (full UI), full-screen mode
- WebSocket-based real-time messaging (fallback: SSE for proxied environments)
- Typing indicators sent within 500ms of first keystroke
- Read receipts (delivered/read double-check marks)
- Online presence indicator (green dot when AI available, amber when human-only)
- File upload: images (JPEG/PNG/WebP <10MB), PDFs (<25MB), stored in S3 with tenant prefix, virus-scanned via ClamAV, served via presigned URLs (24h expiry)
- Emoji picker, formatted text messages (bold, italics)
- Persistent chat history across sessions using `customer_id` cookie or auth token

- Mobile-responsive by default (tested on iOS Safari, Android Chrome)

4 Escalation Scenarios (all P0):

- **Scenario A — Live Transfer:** AI detects frustration or customer requests human → agent pool checked → if agent available (<30s wait), warm transfer with context briefing in agent desktop
- **Scenario B — Callback Request:** No agents available → widget shows callback scheduler → customer selects time → VoiceAgent Engine initiates outbound call at scheduled time
- **Scenario C — Co-Browse:** Agent sends co-browse invitation link → customer accepts → agent sees customer's current browser tab in read-only mode (WebRTC screen sharing via LiveKit)
- **Scenario D — Email Ticket:** Customer selects "Create ticket" or AI determines issue requires async resolution → Email Intelligence Engine creates ticket, sends confirmation email with ticket ID

P1 Requirements:

- Proactive chat triggers: admin configures IF `page_url MATCHES '/pricing'` AND `dwelling_time > 30s` THEN `show_chat({greeting: '...' })`, evaluated client-side; max 1 trigger per session
- Rich message types: image cards, product carousels, quick-reply buttons, forms within chat
- Agent Copilot panel (see Engine 31 below)

Acceptance Criteria:

- Given a tenant adds the `<script>` tag to their website, when the page loads, then the chat widget appears in the configured position within 300ms with no layout shift.
- Given a customer uploads a PDF in chat, when the file is received, then the agent (human or AI) can view the PDF inline and the Document Intelligence Engine indexes it within 60 seconds.
- Given the AI confidence score drops below 0.55 for two consecutive turns, when the AI determines it cannot resolve the query, then it presents the customer with the 4 escalation options within one response turn.
- Given a returning customer opens the chat widget, when they have a previous conversation history, then the last 5 messages are displayed without requiring the customer to re-authenticate on the same device.
- Given an agent is active in the desktop and a chat escalation arrives, when the chat is assigned, then the Agent Copilot panel displays 3 suggested replies within 1 second.

Dependencies: Agent Copilot (Engine 31), 4-Layer Memory, Document Intelligence, Booking Engine, Case Management, Webhook System (chat.message.created events).

Engine 3: WhatsApp Commerce Engine

Problem Solved: WhatsApp is the primary business communication channel in India (500M+ users). Businesses need to conduct full transactional workflows — quoting, booking, payment, and support — within WhatsApp without sending customers to external links.

User Stories:

- As a customer of a travel agency, I want to inquire about a Goa package on WhatsApp and receive a quote without switching apps, so that I can make a buying decision instantly.

- As Rahul (travel agency owner), I want customers to pay a ₹5,000 booking deposit via WhatsApp UPI without leaving the conversation, so that I don't lose bookings at the payment step.
- As an admin, I want to design WhatsApp Flows v2 forms for lead capture and appointment booking, so that my team doesn't have to manually process each inquiry.
- As a customer, I want to receive proactive order status updates on WhatsApp, so that I don't need to call to check status.

P0 Requirements:

- Meta Cloud API integration via AiSensy BSP (approved BSP for India)
- WhatsApp Flows v2 support: dynamic multi-step forms within WhatsApp natively (no web redirect)
- UPI payment collection via WhatsApp Pay / Razorpay link embedded in conversation
- Template message management: create, submit for approval, track approval status in admin portal
- Inbound message handling: text, image, document, audio, location, contacts
- AI-powered response to inbound messages (same LLM stack as ChatAgent)
- Session management: Meta's 24-hour customer-initiated window; template-only outside window
- Opt-in/opt-out management with TRAI-compliant messaging
- Read receipt webhooks from Meta processed into conversation timeline
- Delivery failure handling: WhatsApp → SMS fallback via Exotel for critical notifications

P1 Requirements:

- WhatsApp catalog integration: connect tenant's product/service catalog for in-chat browsing
- Multi-agent WhatsApp inbox with agent assignment and internal notes
- Broadcast campaigns to opted-in customers (see Campaign Engine)

Acceptance Criteria:

- Given a customer sends a WhatsApp message to the business number, when the message is received by the Meta webhook, then the AI responds within 3 seconds.
- Given a customer initiates a booking workflow via WhatsApp Flow, when they submit the form, then a booking draft is created in the Booking Engine and a confirmation message sent within 5 seconds.
- Given a customer sends a UPI payment via WhatsApp Pay, when the payment is confirmed by Razorpay webhook, then the booking is confirmed and a receipt WhatsApp message is sent within 10 seconds.
- Given a customer opts out by replying "STOP", when the opt-out message is received, then the customer is added to the suppression list and no further template messages are sent.

Dependencies: Booking Engine, Invoice Engine, Campaign Engine, Compliance Engine (TRAI opt-out), Payment integration (Razorpay).

Engine 4: Email Intelligence Engine

Problem Solved: Email is the primary channel for formal business communication, complaints, and documentation. It receives the least AI investment despite handling 50%+ of enterprise support volume.

User Stories:

- As Anil (insurance contact center head), I want inbound emails to be automatically classified by intent and urgency, so that my team can focus on high-priority cases first.
- As a customer, I want to receive an immediate auto-acknowledgment with my ticket ID when I email support, so that I know my request was received.
- As an admin, I want to configure SLA timers per email category, so that no email exceeds the promised response time.
- As Kiran (fintech CTO), I want to connect our Gmail workspace to AgentCore via OAuth, so that all inbound support emails are centralized in one system.
- As a customer, I want to click a single "Unsubscribe" link in any email to stop receiving communications, so that I comply with my GDPR and DPDP rights.

P0 Requirements:

- Inbound email ingestion: IMAP IDLE polling or Gmail Push Notifications (Pub/Sub); OAuth 2.0 for Gmail and Outlook; SMTP for custom domains
- Outbound email sending: AWS SES with DKIM + SPF + DMARC configuration per tenant domain
- Email thread grouping: parse In-Reply-To and References RFC 2822 headers; fallback to subject-line matching (strip Re:/Fwd:) + sender pair
- LLM-based classification on receipt: intent (support/billing/sales/complaint/legal), urgency (low/medium/high/critical), sentiment (positive/neutral/negative)
- Auto-routing to agent queues based on classification rules configured in admin portal
- Auto-acknowledgment with ticket ID within 60 seconds of receipt
- SLA timer per category: Starter=24h, Growth=8h, Enterprise=4h first response; escalation at 75% and 90% SLA consumed
- List-Unsubscribe header injected in all outbound marketing/transactional emails
- One-click unsubscribe endpoint (<https://agentcore.ai/unsub/{token}>) with immediate CRM sync
- Bounce handling: SES/SendGrid webhook → hard bounce → suppression list + tenant alert; soft bounce → 3 retries over 24h
- MJML-based HTML email templates with tenant branding (logo, primary color, footer) injected from tenant config; variables: `{{customer.name}}`, `{{ticket.id}}`, `{{agent.name}}`, `{{booking.reference}}`

P1 Requirements:

- AI-generated draft reply suggested to human agent before they open the email (Agent Copilot for email)

- Auto-resolution for common email patterns: "order status" → query booking engine → send automated status reply
- Email thread summarization when handing off between agents
- Attachment handling: PDF/image attachments routed to Document Intelligence Engine for indexing

P2 Requirements:

- Sentiment trend tracking per customer across email history
- Auto-detect email language and respond in matching language

Acceptance Criteria:

- Given an inbound email arrives at the tenant's connected inbox, when the email is received, then it is ingested, classified, and an auto-acknowledgment sent within 60 seconds.
- Given an email thread has the In-Reply-To header set, when a new reply is received, then it is correctly grouped with the existing conversation thread and not treated as a new case.
- Given an email has been in queue for 75% of its SLA duration without a first response, when the SLA alert triggers, then a yellow alert appears in the supervisor dashboard and the assigned agent receives a notification.
- Given a customer clicks the unsubscribe link in an outbound email, when the unsubscribe request is processed, then the customer is added to the suppression list within 5 seconds and the action is logged in the audit trail.

Dependencies: Case Management, Document Intelligence, Agent Copilot, Compliance Engine, Campaign Engine.

Group B — Intelligence Engines

Engine 5: 4-Layer Memory Engine

Problem Solved: AI agents with no memory treat every interaction as the first, frustrating customers and wasting time on re-identification. Persistent, layered memory enables genuine personalization and context-aware service.

Architecture:

- **Layer 1 — Redis Working Memory:** Active session state. TTL: 24 hours. Stores: current conversation turns, detected intent, active workflow state, pending tool calls. Per-customer lock via Redis SET NX EX for concurrent channel writes.
- **Layer 2 — PostgreSQL Episodic Memory:** All conversation transcripts, interaction history, agent notes. Indexed by customer_id, tenant_id, channel, created_at. Partitioned monthly. Default retention: 90 days (configurable per tenant). GDPR erasure: DELETE WHERE customer_id = X cascades to all tables.
- **Layer 3 — pgvector Semantic Memory:** Embeddings generated by OpenAI text-embedding-3-small (1536 dimensions). One embedding per conversation summary. IVFFlat index (100 lists) for <1M vectors per tenant; HNSW for >1M. Used for "have we heard this complaint before?" and customer preference retrieval.

- **Layer 4 — PostgreSQL Procedural Memory:** Tenant-specific business rules, compliance scripts, escalation logic, SLA thresholds, agent routing rules. Immutable during conversations; updated only via admin portal.

PO Requirements:

- On every new conversation: load customer's episodic memory (last 5 interactions) + semantic retrieval (top 3 relevant past conversations) + procedural rules for tenant
- Memory consolidation job: every 30 minutes, compress Redis working memory into PostgreSQL episodic record
- Memory size limits per tier: Starter=10K conversations, Growth=100K, Enterprise=unlimited; soft limit alert at 80%, hard limit archives oldest to S3 Glacier
- Data export API: GET /api/v1/tenants/{id}/memory/export returning JSON/CSV; async job with email download link; rate-limited to 1 export/hour
- Customer-level data deletion: DELETE /api/v1/customers/{id}/memory purges all layers within 72 hours (GDPR/DPDP compliance)

Acceptance Criteria :

- Given a returning customer calls for the second time, when the call is connected, then the voice agent opens the conversation with a reference to the previous interaction (e.g., "Welcome back, Ramesh. Last time we spoke about your Manali booking—") within the first 5 seconds.
- Given a customer requests deletion of their data, when the deletion API is called, then all memory layers return zero records for that customer within 72 hours.

Dependencies: VoiceAgent, ChatAgent, WhatsApp Engine, all CRM Lifecycle Engines.

Engine 6: Multi-Agent Orchestrator

Problem Solved: Complex customer requests require multiple specialized agents working in parallel. A LangGraph Supervisor pattern coordinates 7 parallel workers per conversation without sequential bottlenecks.

7 Workers:

1. **Intent Router:** Classifies intent from first message; routes to relevant engines; confidence score drives escalation
2. **Memory Retrieval Agent:** Queries all 4 memory layers in parallel; packages context for Supervisor within 500ms
3. **Tool Execution Agent:** Executes CRM API calls, booking lookups, payment processing; manages tool timeouts (max 10s per tool, 15s for payment)
4. **Language/Translation Agent:** Detects language, translates if needed, selects appropriate STT/TTS stack
5. **Compliance Agent:** Validates each AI response against tenant's compliance rules and TRAI/DPDP requirements before sending
6. **Proactive Suggestion Agent:** Monitors conversation for upsell/cross-sell triggers; pushes suggestions to Supervisor for injection

7. **Quality/Escalation Agent:** Tracks sentiment trajectory, resolution confidence, handle time; triggers escalation when thresholds breached

P0 Requirements:

- LangGraph Supervisor process is stateless; all state stored in Redis (resilience to supervisor crash)
- If supervisor crashes, Kubernetes liveness probe restarts within 5 seconds; new supervisor reads Redis state and resumes
- Per-engine timeouts: Understanding engines (Intent, Language) = 2s; Action engines (Tool Execution) = 10s; Payment = 15s; Compliance = 1s
- Circuit breaker per engine: 5 failures within 60s → skip engine, return graceful degradation response; 60s cooldown before retry
- Engine versioning: semver per engine; Supervisor routes to version via tenant feature flag; canary deployment: 5% traffic → monitor → promote
- OpenTelemetry integration: every engine emits spans with {engine_id, tenant_id, latency_ms, status, request_id}

Acceptance Criteria:

- Given 7 parallel workers execute for a single conversation turn, when all workers complete, then the Supervisor assembles the response within 2 seconds of the customer's message.
- Given the Tool Execution Agent makes 3 consecutive failed API calls to an external CRM within 30 seconds, when the circuit breaker fires, then a graceful degradation response is returned and an alert is sent to the tenant admin.

Dependencies: All other engines; Redis; LangGraph; OpenTelemetry.

Engine 7: Proactive Outbound Engine

Problem Solved: Most customer relationship value is lost in the spaces between interactions — when follow-up doesn't happen, reminders aren't sent, and reactivation is forgotten.

P0 Requirements:

- TRAI DND registry check before every outbound call/SMS; automatic skip and log for DND numbers
- Scheduled call triggering: admin defines rules (e.g., "Call lead if no reply to quote after 48 hours")
- Outbound WhatsApp template messages for appointment reminders, payment reminders, booking confirmations
- Campaign scheduling with time-window rules (only call between 9am–8pm IST)
- Retry logic: up to 3 attempts with configurable intervals; escalate to human queue after 3 failed attempts
- Full audit log: every outbound attempt logged with timestamp, outcome (answered/not answered/DND), duration

Acceptance Criteria:

- Given a lead's proposal was sent 48 hours ago with no response, when the follow-up rule triggers, then an outbound call is initiated and the voice agent opens with the proposal context already loaded.
- Given a customer's number is on the TRAI DND registry, when the outbound engine checks before dialing, then the call is skipped, the skip is logged, and no call is placed.

Dependencies: VoiceAgent Engine, WhatsApp Commerce Engine, Lead Engine, Compliance Engine.

Group C — CRM Lifecycle Engines

Engine 8: Lead Engine

Problem Solved: Leads arrive from multiple channels (voice, chat, WhatsApp, web forms, social DMs) and are lost in disconnected systems. Systematic capture, scoring, and routing is required.

P0 Requirements:

- Unified lead inbox: captures from voice, chat, WhatsApp, web form, email, social DMs
- Auto-creation of lead record from any inbound inquiry with: name, phone, email, channel, source, intent, lead_score, assigned_agent, created_at
- Lead scoring engine: rule-based (intent strength, response time, budget signals) + ML-based (conversion probability)
- Auto-assignment: round-robin or skill-based routing to available agents
- Duplicate detection: merge leads with matching phone or email within same tenant
- Lead stage tracking: New → Contacted → Qualified → Proposal Sent → Negotiating → Won/Lost
- WhatsApp/email notification to assigned agent on new lead within 30 seconds

Acceptance Criteria:

- Given a customer calls the business and the AI captures their name, phone, and travel interest, when the call ends, then a lead record is created within 10 seconds with the captured data pre-populated.

Dependencies: VoiceAgent, ChatAgent, WhatsApp Engine, Email Intelligence, Proposal Engine.

Engine 9: Proposal Engine

Problem Solved: Proposal creation is time-consuming and inconsistent. AI should generate customized proposals from conversation data and business knowledge.

P0 Requirements:

- Template-based proposal generation from conversation data
- Dynamic content blocks: itinerary details, pricing tables, terms and conditions, agent bio
- Multi-format output: PDF (via WeasyPrint) and WhatsApp-formatted text
- E-signature integration: DocuSign or Adobe Sign via API
- Proposal status tracking: Draft → Sent → Opened → Accepted/Rejected

- Tenant branding: logo, colors, fonts injected from tenant config

Acceptance Criteria:

- Given a lead conversation captures destination, dates, and party size, when the Proposal Engine is triggered, then a formatted PDF proposal is generated within 30 seconds and sent via email and WhatsApp simultaneously.

Engine 10: Quote Engine

Problem Solved: Real-time pricing quotes require accurate inventory, pricing rules, and margin calculations — previously a manual, error-prone task.

P0 Requirements:

- Pricing rule engine: base price + modifiers (season, pax count, lead time, promotion code)
- Supplier price lookup integration (APIs or manual price sheets loaded via CSV)
- Margin calculation with minimum margin guardrails (reject quotes below configured threshold)
- Quote versioning: customer can see quote history, agent can see what was sent
- Quote-to-booking conversion in one click
- Validity period configuration (default: 48 hours; configurable per tenant)

Engine 11: Case Management Engine

Problem Solved: Customer complaints and issues require structured tracking, SLA enforcement, and resolution workflows.

P0 Requirements:

- Case creation from any channel: voice, chat, WhatsApp, email, agent manual entry
- Case types: complaint, refund request, escalation, feedback, billing dispute, technical issue
- Priority levels: P1 (critical, <4h), P2 (high, <8h), P3 (medium, <24h), P4 (low, <72h)
- SLA timer with visual escalation: green → yellow (75%) → red (90%) → breached
- Case assignment: manual or auto-assign based on skill and availability
- Internal notes (agent-only) vs. customer-visible updates
- Case merge: duplicate cases for same customer/issue linked and merged
- Case resolution templates: pre-written resolution messages per category
- Reopen window: customer can reopen within 7 days of resolution

Acceptance Criteria:

- Given a customer emails a complaint, when the Email Intelligence Engine classifies it as a complaint with high urgency, then a P2 case is auto-created, assigned to the compliance-trained agent queue, and an acknowledgment email sent within 60 seconds.

Engine 12: Approval/Workflow Engine

Problem Solved: Multi-step business processes (refund approvals, discount authorizations, booking modifications) require structured approvals with role-based authorization.

P0 Requirements:

- Visual workflow builder: drag-and-drop nodes (trigger, action, condition, approval, notification)
- Approval nodes: single Approver or multi Approver (any/all)
- Escalation on approval timeout: escalate to next Approver after configurable time
- Role-based approval authority: define maximum approval amounts per role
- Audit trail: every workflow step logged with actor, timestamp, decision, comment
- Workflow templates for common use cases: refund approval, discount authorization, booking modification, leave request

Engine 13: Booking Engine

Problem Solved: AI agents must be able to create, modify, and cancel real bookings — not just collect booking intent and hand off to humans.

P0 Requirements:

- Booking creation from voice/chat/WhatsApp conversations
- Availability checking against real inventory (supplier API or manual availability calendar)
- Booking confirmation: unique booking reference, confirmation via WhatsApp + email
- Modification workflow: customer requests change → availability check → price recalculation → confirmation
- Cancellation with policy enforcement (configured cancellation rules per tenant)
- Double-booking prevention: atomic booking creation with pessimistic locking
- Booking calendar view in admin portal

Acceptance Criteria :

- Given a customer requests a hotel booking for specific dates via WhatsApp, when the Booking Engine confirms availability, then the booking is created, a reference number is assigned, and confirmation is sent via WhatsApp within 10 seconds.

Engine 14: Invoice Engine

Problem Solved: Manual invoice generation is slow, error-prone, and creates accounting overhead.

P0 Requirements:

- Auto-invoice generation on booking confirmation
- GST-compliant invoice format (CGST/SGST/IGST based on customer state)
- Invoice numbering with sequential, tenant-namespaced format

- PDF generation (WeasyPrint) and email delivery
- Payment link embedding (Razorpay payment gateway)
- Invoice status: Draft → Sent → Paid → Overdue → Cancelled
- Overdue alerts: D+7, D+14, D+30 reminders via WhatsApp and email

Engine 15: Refund/Cancellation Engine

Problem Solved: Refund processing is a major customer pain point and compliance risk. Inconsistent handling generates complaints and regulatory exposure.

P0 Requirements:

- Refund policy engine: configurable rules (e.g., "100% refund if cancelled 15+ days before travel, 50% if 7-14 days, 0% within 7 days")
- Automated eligibility check on refund request
- Razorpay refund API integration for payment reversal
- Refund status tracking: Requested → Approved → Processing → Completed
- Refund timeline commitments: standard 5-7 banking days; express 1-2 days
- Case creation for every refund request (linked to Case Management Engine)
- Refund audit trail: full log of every state change with actor and timestamp

Group D — Operations Engines

Engine 16: Document Intelligence Engine

Problem Solved: Businesses receive critical information in unstructured documents (passports, invoices, medical records, contracts). AI must extract, validate, and store structured data.

P0 Requirements:

- Document ingestion: PDF, JPEG, PNG, DOCX, XLSX, CSV via upload or email attachment
- OCR via AWS Textract for scanned documents
- LLM-based information extraction: structured JSON from unstructured text
- Document classification: identity documents, invoices, medical records, contracts, receipts
- Validation rules per document type (e.g., passport expiry must be >6 months from travel date)
- Extracted fields stored in customer profile with source document reference
- Document storage: S3 with tenant-scoped prefix, AES-256 encryption at rest

Acceptance Criteria:

- Given a customer uploads a passport image in the chat widget, when Document Intelligence Engine processes it, then the customer's name, nationality, passport number, and expiry date are extracted and stored in their profile within 60 seconds.

Engine 17: Analytics Engine

Problem Solved: Businesses need to understand AI performance, agent productivity, and customer satisfaction trends to continuously improve.

P0 Requirements:

- Pre-built dashboards: Conversation Volume, AI Resolution Rate, Escalation Rate, Average Handle Time, CSAT, NPS
- Date range filtering: today, last 7 days, last 30 days, custom range
- Channel breakdown: voice vs. chat vs. WhatsApp vs. email
- Agent performance report: conversations handled, CSAT per agent, AHT per agent, resolution rate
- Peak hour heatmap: conversation volume by hour and day of week
- Query builder: custom metrics from any stored conversation/interaction field
- Data export: CSV download for any report
- RDS read replica for analytics queries (no impact on primary DB performance)

P1 Requirements:

- AI-generated weekly summary: natural language summary of key trends emailed to tenant admin Monday morning
- Anomaly detection: alert on unusual spike or drop in any KPI ($>2\sigma$ deviation from 30-day average)
- Funnel analysis: Lead → Proposal → Quote → Booking conversion rates per channel and period

Engine 18: Supplier Management Engine

Problem Solved: Businesses in travel, logistics, and hospitality manage dozens of suppliers. AI needs access to supplier pricing, availability, and contact data to execute transactions.

P0 Requirements:

- Supplier profiles: name, type, contact, SLA, payment terms, API credentials
- Supplier API integration framework: REST webhook from supplier → normalized internal format
- Manual price sheet upload (CSV/XLSX) with automatic parsing
- Supplier availability calendar: block dates, capacity limits
- Supplier performance tracking: on-time delivery rate, cancellation rate, average response time

Engine 19: Campaign Engine

Problem Solved: Businesses need to send targeted, personalized outbound communications to opted-in customer segments.

P0 Requirements:

- Segment builder: filter customers by demographics, purchase history, tag, location, last interaction date
- Campaign types: WhatsApp broadcast, email newsletter, voice blast (TRAI-compliant)
- Template selection with personalization variables
- Send-time optimization: schedule based on customer's past engagement patterns
- Campaign analytics: sent, delivered, opened, clicked, replied, opted out
- Opt-out honoring: auto-suppress any customer who opts out during a campaign
- TRAI DND check before every outbound campaign contact

Engine 20: NPS/Feedback Engine

Problem Solved: Systematic customer feedback collection is the primary mechanism for product improvement and relationship health monitoring.

PO Requirements:

- Post-interaction survey trigger: auto-send after voice call, chat session, or booking completion
- Survey channels: WhatsApp (5-star + optional text), email (full form), SMS
- NPS calculation: standard promoters/passives/detractors scoring
- CSAT and CES (Customer Effort Score) options
- Response tagging: auto-categorize open-text responses by topic (speed, staff, pricing, resolution)
- Trend dashboard: NPS over time, breakdown by channel, agent, and interaction type
- Alert on negative feedback: auto-create case and assign to retention team when CSAT ≤ 2

Engine 21: Compliance Engine

Problem Solved: Regulatory non-compliance in AI-driven communications generates legal liability, fines, and reputational damage.

PO Requirements:

- TRAI DND check: API integration with NDNC registry; check before every outbound call and SMS
- TRAI disclosure: configurable mandatory disclosure text played before every AI voice call ("This call is from [Business Name]. It may be recorded for quality purposes.")
- DPDP consent capture: explicit consent logged before data collection; stored with timestamp, channel, and text of consent provided
- DPDP erasure: deletion API that cascades across all memory layers within 72 hours
- HIPAA mode: when enabled for healthcare tenants, enforces: no PHI in logs, BAA requirement, encryption verification, access controls with audit trail
- PCI compliance: no full card numbers stored in any log; Razorpay tokenization enforced; SAQ-A self-attestation documentation

- Per-jurisdiction recording rules: engine blocks recording in two-party consent states/regions unless consent is captured
- Compliance audit log: immutable log of every compliance event (consent capture, DND check, disclosure played, erasure executed)

Engine 22: Tenant Management Engine

Problem Solved: A SaaS platform serving thousands of tenants — including white-label partners with their own sub-tenants — requires a robust, hierarchical tenancy system.

Three-Level Hierarchy:

- **Level 0 — Platform (HelpRevX):** Full visibility across all tenants; billing admin; compliance override
- **Level 1 — Partner (White-Label Reseller):** Creates and manages their sub-tenants; sees aggregated data across their clients; billed for all sub-tenant usage; cannot see other partners' data
- **Level 2 — End Client (Tenant):** Sees only their own data; configures their own agents, workflows, and team

P0 Requirements:

- `parent_tenant_id` nullable FK in `tenants` table; max 3-level hierarchy (Platform → Partner → End Client)
- Hierarchical Row Level Security (RLS) in PostgreSQL: End Client sees `tenant_id = their_id`; Partner sees `tenant_id = their_id OR parent_tenant_id = their_id`; Platform sees all
- Tenant provisioning API: `POST /api/v1/tenants` → auto-provisions RLS policies, creates default config, assigns subdomain, sends welcome email; state machine: `PENDING` → `ACTIVE` → `SUSPENDED` → `DELETED`
- Tenant feature flags: `tenant_features` table with `{tenant_id, feature_key, enabled, config_json}`; cached lookup (Redis TTL 60s); integrated with LaunchDarkly or Unleash
- Billing rollup: partner invoiced for sum of all sub-tenants' usage; sub-tenant reports show individual usage; partner sees all sub-tenant usage breakdown
- Tenant isolation test suite: automated test runs on every deploy; creates 2 tenants, inserts data, queries as tenant A, asserts zero rows from tenant B

Sub-Tenant Architecture Detail:

```
CREATE TABLE tenants (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  parent_tenant_id UUID REFERENCES tenants(id) NULL,
  name TEXT NOT NULL,
  plan ENUM('starter', 'growth', 'scale', 'enterprise', 'white_label') NOT NULL,
  status ENUM('pending', 'active', 'suspended', 'deleted') DEFAULT 'pending',
  subdomain TEXT UNIQUE,
  created_at TIMESTAMPTZ DEFAULT now()
);
-- RLS policy
CREATE POLICY tenant_isolation ON conversations
```

```
USING (
  tenant_id = current_setting('app.current_tenant_id')::uuid
OR tenant_id IN (
  SELECT id FROM tenants WHERE parent_tenant_id = current_setting('app.current_tenant_id')
)
);
```

Acceptance Criteria:

- Given Partner A has sub-tenants C1 and C2, when Partner A queries the analytics dashboard, then they see aggregated data for C1 and C2 but no data from any other partner's tenants.
- Given End Client C1 makes an API call with their API key, when querying conversations, then only conversations with `tenant_id = C1.id` are returned, with zero rows from any other tenant.

Engine 23: Itinerary Builder Engine

Problem Solved: Travel agencies need to create structured, multi-day itineraries dynamically from conversation data and supplier inventory.

P0 Requirements:

- Day-by-day itinerary builder: add/remove/reorder activities, hotels, transfers
- AI-powered itinerary generation: from conversation ("5 days in Rajasthan, couple, honeymoon, budget ₹80,000") → draft itinerary within 60 seconds
- Itinerary PDF export with agency branding
- Supplier price lookup per activity/hotel in itinerary
- Itinerary sharing: unique link sent to customer for review and approval
- Version history: track all modifications with agent name and timestamp

Engine 24: Expense Capture Engine

Problem Solved: Field staff and business travelers need to capture expenses via voice and WhatsApp without manual data entry, connecting to SpendPilot for approval and reimbursement.

P0 Requirements:

- Voice expense capture: "Spent 450 on auto from airport" → parses amount, category, merchant from natural language
- WhatsApp expense: photo of receipt → Document Intelligence Engine extracts amount, merchant, date → creates expense record
- Expense categorization: auto-classify by vendor name and description
- Policy compliance check: flag expenses that violate tenant's expense policy (e.g., meals >₹500/day)
- SpendPilot integration: push approved expenses to SpendPilot via API for reimbursement
- Monthly expense report generation per employee

Cross-Cutting Requirements

Engine 25: Chat Widget & Embed System *(see Engine 2 above for full spec)*

Additional technical requirements:

- Widget bundle size: <50KB gzipped for initial load; agent logic lazy-loaded
- CSP header compatibility: `script-src cdn.agentcore.ai; frame-src cdn.agentcore.ai`
- CloudFront CDN: `cdn.agentcore.ai` served from CloudFront with edge caching; widget JS versioned with content hash for cache busting
- Performance: widget initialization completes within 300ms on a 4G connection
- Graceful degradation: if CDN fails, widget silently fails without breaking host page

Engine 26: Mobile SDK

P0 Requirements:

- iOS SDK: Swift Package Manager distribution; supports iOS 14+; `AgentCore.startVoiceSession()`, `AgentCore.startChatSession()`, `AgentCore.endSession()` APIs
- Android SDK: Maven/Gradle distribution; supports Android API 21+; Kotlin-first with Java compatibility
- React Native Bridge: npm package `@agentcore/react-native`; wraps native iOS/Android SDKs
- Flutter Plugin: pub.dev package `agentcore_flutter`
- Push notification registration: FCM (Android) and APNs (iOS); AgentCore sends push via Firebase Admin SDK when proactive message available
- SDK event callbacks: `onConversationStarted`, `onMessageReceived`, `onAgentHandoff`, `onConversationEnded`
- SDK documentation: API reference + quick start guide per platform; CocoaPods support for iOS

Engine 27: Webhook System

P0 Requirements:

- Webhook subscription management: `POST /api/v1/webhooks` with `{url, events[], secret}`
- HMAC-SHA256 signature: `X-AgentCore-Signature: t=TIMESTAMP, v1=HASH`; payload = raw request body; key = per-tenant webhook secret (32-char, rotatable via portal); reject if timestamp >5 minutes old (replay attack prevention)
- Event catalog (50+ events): `call.started`, `call.ended`, `call.transferred`, `call.recording.ready`, `chat.session.started`, `chat.message.received`, `chat.escalated`, `lead.created`, `lead.qualified`, `booking.created`, `booking.confirmed`, `booking.cancelled`, `invoice.created`, `invoice.paid`, `refund.requested`, `refund.completed`, `case.created`, `case.resolved`, `nps.received`, `campaign.sent`, `document.processed`, `tenant.provisioned`, `agent.logged_in`
- Delivery with retry: initial attempt within 1s; retry on 4xx/5xx or timeout: exponential backoff (2s, 4s, 8s, 16s, 30s max); 5 attempts total; after 5 failures, webhook marked failed and tenant

notified

- Webhook dashboard in admin portal: each event log with status (delivered/failed), response code, latency, payload preview, manual retry button
- Webhook test button: admin can send a test payload for each event type

Engine 28: Billing & Metering API

P0 Requirements:

- Usage metering: real-time event tracking for: voice minutes (rounded up to nearest second), WhatsApp messages (template vs. session), API calls (per 1,000 requests), active tenants
- Metering pipeline: usage events → Redis counter → 1-minute batch flush → PostgreSQL `usage_events` table
- Invoice generation: automated on the 1st of each month; PDF invoice with line items per usage category; delivered via email and available in admin portal
- Razorpay integration for Indian tenants: auto-charge saved card on invoice due date
- Stripe integration for international tenants: SCA-compliant payment flows
- Usage alerts: notification at 80% of plan allowance (in-app + email); at 100%, tenant can continue at overage rates (configurable) or services pause until top-up
- Usage dashboard: tenant sees real-time usage for current billing period with projected overage
- Sub-tenant billing: partner sees aggregated usage across all their sub-tenants; breakdown by sub-tenant available

Engine 29: Sub-Tenant Architecture *(detailed in Engine 22 above)*

Engine 30: Rate Limiting & Noisy Neighbor Protection

P0 Requirements:

- Per-tenant token bucket for API requests (Redis Lua script for atomic check-and-decrement)
 - Starter: 100 requests/min; Growth: 500 requests/min; Scale: 2,000 requests/min; Enterprise: custom
- Per-tenant concurrent call limit enforced in LiveKit media server: Starter=5, Growth=20, Scale=50, Enterprise=200
- Burst protection: 2× burst allowance for 10-second windows; sustained burst triggers graceful degradation (HTTP 429 with `Retry-After` header)
- Per-tenant CPU/memory cgroup limits on ECS tasks: prevent single tenant consuming >20% of cluster capacity
- Fair-use queuing: when cluster is at capacity, new requests enter per-tenant queues; FIFO within tenant, round-robin across tenants; max queue depth 100 requests per tenant

Engine 31: Agent Copilot (AI Assist for Human Agents)

P0 Requirements:

- Real-time suggested replies: on every customer message, generate top 3 suggested agent responses ranked by confidence score and relevance; update within 1 second of customer message receipt; agent clicks to insert (editable before send)
- KB article suggestion panel: semantic search of knowledge base on each customer message; top 3 articles displayed with relevance score and excerpt; expandable in-panel
- Customer 360 sidebar: displays customer lifetime value, interaction history (last 10), open cases, sentiment trend, churn score, purchase history, tags
- Next-best-action recommendations: based on conversation context, display 1–3 suggested actions ("Offer loyalty discount — customer in top 20% by spend", "Escalate to manager — third complaint this month")
- Tone coaching: if agent draft reply contains negative or unprofessional language, surface a warning with rewrite suggestion
- Auto-draft reply generation: single click generates a complete, professional reply draft based on context; agent edits and sends

Acceptance Criteria:

- Given an agent receives a customer chat, when the customer sends a message, then the Copilot panel refreshes with 3 new suggested replies and up to 3 relevant KB articles within 1 second.
- Given an agent types a reply containing flagged language (e.g., "That's not my problem"), when the tone checker detects the issue, then a yellow warning banner appears with a suggested rewrite before the agent can send.

Engine 32: Design System Specification

Color Tokens:

```
--navy: #1a1f36;      /* Primary background, headings */
--accent: #4f8ef7;     /* CTAs, links, active states */
--success: #22c55e;    /* Confirmed, resolved, online */
--warning: #f59e0b;    /* SLA at risk, usage alerts */
--error: #ef4444;      /* Failed, escalated, offline */
--surface: #ffffff;    /* Card backgrounds */
--muted: #f4f5f7;     /* Page backgrounds */
--text-primary: #1a1f36;
--text-secondary: #6b7280;
```

Typography Scale:

- Display: 32px / 40px line-height / 700 weight — for hero headings
- H1: 24px / 32px / 700 — for page titles
- H2: 20px / 28px / 600 — for section headers
- H3: 16px / 24px / 600 — for card headers

- Body: 14px / 20px / 400 — for all body text
- Caption: 12px / 16px / 400 — for helper text, timestamps

Component Library:

- Base: Shadcn/UI (Radix UI primitives + Tailwind CSS)
- Custom components: ConversationTimeline, AgentCard, MetricCard, VoiceWaveform, StatusBadge, TenantSwitcher
- Dark mode: full dark mode support via Tailwind dark: classes; user preference stored in localStorage and synced to user profile
- WCAG 2.1 AA: all interactive elements have $\geq 4.5:1$ contrast ratio; all form inputs have visible labels; all icons have aria-labels; full keyboard navigation with visible focus states

Engine 33: Onboarding Wizard

P0 Requirements — 5-Step Wizard:

1. **Industry Selection:** 15 industry cards with icons; selection pre-loads industry-specific templates (voice persona, FAQ knowledge base, escalation rules)
2. **Voice Persona Configuration:** Choose name, gender, accent (Indian English / Hindi / regional), greeting message, hold music; play sample button
3. **Business Hours Setup:** Visual time-picker for operating hours per day; timezone selector; after-hours behavior (voicemail / AI-only / AI + callback offer)
4. **First Tool Registration:** Connect first integration (Exotel phone number, WhatsApp Business number, or embed chat widget); guided walkthrough with screenshots
5. **Test Call / Test Chat:** Trigger a live test of the configured AI agent; pass/fail validation with specific error messages

Target: New tenant completes wizard and has a working AI agent in under 30 minutes.

P0 Requirements:

- All 15 industry templates pre-loaded (travel, photography, healthcare, BFSI, real estate, education, fitness, automotive, hospitality, retail, logistics, home services, events, legal, NGO)
- Progress indicator (1 of 5 steps, estimated time remaining)
- "Skip and do later" option for each step except Step 4 (at least one channel required)
- Save progress: wizard state saved to backend; user can resume on next login
- Completion event triggers onboarding email sequence (Day 1, Day 3, Day 7 check-in emails)

PART 7: USER STORIES (EXPANDED)

Voice Experience

1. As a first-time caller to a travel agency, I want to hear a natural-sounding AI agent greet me by name if I'm a returning customer, so that I feel recognized rather than treated as a stranger.
2. As a patient calling a hospital after hours, I want the AI to tell me the emergency contact number and take my appointment request for the next day, so that I'm not left without assistance.
3. As a customer calling during the AI's business hours, I want the AI to verify my identity using my registered mobile number, so that my account information is protected.
4. As a customer with a complaint, I want to be transferred to a human supervisor immediately when I say "I want to speak to a manager," so that my escalation is taken seriously.
5. As a caller who speaks Tamil, I want the AI to detect my language and respond in Tamil without me having to press any language selection key, so that the experience feels natural.
6. As a customer, I want to receive a WhatsApp summary of my call — including any actions the agent took — within 5 minutes of hanging up, so that I have a record.
7. As a customer making a payment query, I want the AI to read me my outstanding balance and offer a payment link via SMS/WhatsApp, so that I can pay immediately without calling back.
8. As an admin, I want calls that exceed 15 minutes to automatically alert a supervisor, so that long-handle-time calls get attention.
9. As a customer, I want to hear clear hold music with periodic updates ("You're next in queue, estimated wait 2 minutes") rather than dead silence, so that I don't hang up.
10. As a customer, I want to opt out of call recording by pressing 9 when prompted, so that I can exercise my privacy rights without penalty.

Chat Experience

11. As a website visitor, I want the chat widget to load without slowing down the host page, so that my browsing experience is unaffected.
12. As a customer seeking support, I want to send a screenshot of my issue in the chat, so that the agent doesn't need to ask me to describe the problem.
13. As a customer starting a new chat, I want to see the name and photo of my assigned agent (or the AI persona's avatar), so that the conversation feels human and personal.
14. As an agent receiving a chat, I want to see the customer's complete history and all open cases in a sidebar, so that I have full context before responding.
15. As a customer on a pricing page, I want a proactive chat pop-up after 30 seconds asking if I have questions about pricing, so that I get help at the moment I need it.
16. As a customer waiting for an agent, I want to see a real-time queue position ("You're #3 in queue") rather than a generic "please wait" message, so that I can decide whether to wait or request a callback.
17. As a customer, I want to receive a WhatsApp notification when my chat session is about to time out due to inactivity, so that I can respond and keep the session open.
18. As an agent, I want to initiate co-browsing with a customer who is confused about a form, so that I can guide them without requiring them to describe what they see.

19. As an admin, I want to set maximum chat queue size (e.g., 10 waiting customers), so that when the queue is full, new customers are offered callbacks rather than waiting indefinitely.
20. As a customer chatting from a mobile device, I want the chat widget to use the full screen in mobile mode and support swipe-to-close, so that it feels like a native app.

WhatsApp Experience

21. As a customer, I want to receive my booking confirmation on WhatsApp with a PDF attachment, so that I have the document in a familiar app.
22. As a customer, I want to use a WhatsApp Flow to book an appointment without typing, so that the process is fast and error-free on mobile.
23. As a travel customer, I want to pay my booking deposit via WhatsApp Pay without being redirected to a browser, so that the payment is completed in one interaction.
24. As an admin, I want to preview a WhatsApp template message before submitting it to Meta for approval, so that I can catch formatting errors.
25. As a customer, I want to reply STOP to unsubscribe from all marketing messages with immediate effect, so that I can exercise my opt-out right instantly.
26. As an admin, I want to schedule a WhatsApp reminder campaign for all customers with appointments in the next 24 hours, automatically, so that no-show rates are reduced.
27. As a customer, I want to receive a WhatsApp message with my refund status update rather than having to check a portal, so that I'm informed proactively.
28. As an agent, I want to take over a WhatsApp conversation from the AI when the customer asks for a human, so that the handoff is seamless and the chat history is visible.

Agent Experience

29. As a human agent, I want to see all channels (phone, chat, WhatsApp, email) in a single unified inbox, so that I'm never jumping between tools.
30. As an agent, I want to set my status (available, busy, on break, offline) and have the routing system respect my status, so that I'm not assigned conversations when I'm unavailable.
31. As an agent handling a complaint, I want a pre-written resolution script for common complaint types to appear automatically, so that I follow the correct protocol without memorizing it.
32. As a supervisor, I want to see a real-time dashboard of all active conversations with color-coded SLA status, so that I can intervene before breaches occur.
33. As a supervisor, I want to whisper coaching tips to an agent during a live call without the customer hearing, so that I can guide without interrupting.
34. As an agent, I want to create an internal note on a case visible only to my team, so that context is shared without the customer seeing internal discussions.
35. As an agent, I want the Copilot panel to suggest a discount offer when the customer's churn risk score is high, so that I can make a retention offer proactively.
36. As an agent, I want to transfer a WhatsApp conversation to a colleague with a briefing note, so that the receiving agent starts with full context.

- 37. As an agent going off-shift, I want all my open conversations to be automatically reassigned to the on-shift queue, so that no customer is left waiting.
- 38. As an agent, I want to see the AI-generated summary of a long email thread condensed to 3 bullet points, so that I can respond quickly without reading 20 back-and-forth messages.

Admin Experience

- 39. As an admin, I want to create and publish a new knowledge base article from a resolved conversation in one click, so that the AI learns from real customer interactions.
- 40. As an admin, I want to configure different AI personas for different departments (friendly persona for sales, formal persona for legal/compliance), so that each interaction matches the context.
- 41. As an admin, I want to receive a weekly analytics digest by email every Monday summarizing key KPIs, so that I don't need to log in to see the big picture.
- 42. As an admin, I want to configure business hours and holiday schedules once and have them apply across all channels, so that the AI never tells customers it's open when it isn't.
- 43. As an admin, I want to view a complete audit trail of every action taken by the AI agent (calls placed, bookings created, refunds processed), so that I can audit AI behavior without reading transcripts.
- 44. As an admin, I want to set approval workflows for refunds above ₹10,000, so that large refunds require manager sign-off.
- 45. As an admin, I want to configure the email signature and branding (logo, colors) once and have it apply to all outbound emails and chat widgets automatically.

Developer Experience

- 46. As a developer integrating AgentCore via Model A API, I want a comprehensive OpenAPI 3.1 spec and interactive documentation, so that I can integrate without a sales call or support ticket.
- 47. As a developer, I want webhook events to include HMAC-SHA256 signatures, so that I can verify authenticity before processing in my system.
- 48. As a developer, I want to provision a new tenant programmatically via API in a CI/CD pipeline, so that I can automate test environment setup.
- 49. As a developer, I want API error responses to include a `request_id`, error code, and human-readable message, so that I can debug integration issues without contacting support.
- 50. As a developer, I want to test webhook delivery in the admin portal without triggering a real event, so that I can validate my webhook handler during development.
- 51. As a developer, I want to use cursor-based pagination on all list endpoints, so that I can efficiently page through large datasets without offset drift.
- 52. As a developer building a React Native app, I want an official React Native bridge SDK with TypeScript types, so that I can add AI voice to my app without bridging native code myself.

PART 8: ACCEPTANCE CRITERIA (10 Critical Features)

AC1: VoiceAgent Engine — End-to-End Call Handling

1. Given a new inbound call arrives, When LiveKit receives the WebRTC signal, Then the AI agent must answer and play the opening greeting within 2,000ms of the first ring.
2. Given the AI agent is speaking, When the customer starts speaking (VAD detects activity), Then the AI must stop within 200ms (barge-in) and begin processing the customer's speech.
3. Given a customer speaks in Hindi after an English greeting, When language detection confidence exceeds 0.85, Then the AI switches all subsequent TTS to Sarvam Bulbul v3 and STT to Sarvam Saaras V3 within the next response turn.
4. Given the AI produces a response in cascaded mode, When total pipeline latency (STT+LLM+TTS) exceeds 1,200ms, Then the system attempts S2S mode on the next turn with a latency target of <400ms.
5. Given a call ends for any reason (customer hangup, transfer, error), When the call terminates, Then a transcript, call summary, and metadata record are created in the database within 30 seconds.
6. Given a customer explicitly says "don't record this call," When the phrase is detected, Then call recording stops within 1 second, a confirmation message plays, and the opt-out is logged with timestamp.
7. Given the Compliance Engine has TRAI mode enabled, When any outbound call is initiated, Then the system queries the NDNC registry first, and if the number is listed, the call is blocked and the skip is logged.

AC2: Chat Widget Embed

1. Given a tenant adds the AgentCore `<script>` tag to a webpage, When the page loads on a standard 4G connection, Then the chat bubble appears within 300ms without any visible layout shift.
2. Given the host page uses a strict Content Security Policy, When the widget loads, Then all widget resources are served from `cdn.agentcore.ai` and no CSP violation is thrown.
3. Given a customer closes the chat window and reopens it within the same browser session, When the widget reinitializes, Then the full previous conversation is visible without requiring any re-authentication.
4. Given a customer sends a file attachment (PDF, <25MB), When the upload completes, Then the file is virus-scanned within 10 seconds, stored in S3, and a presigned link valid for 24 hours is available to the agent.
5. Given the widget is viewed on a mobile device (320px–414px width), When the chat is expanded, Then the widget fills the full viewport and all interactive elements are touch-accessible with minimum 44×44px tap targets.

AC3: 4-Layer Memory Engine — Cross-Session Context

1. **Given** a customer has had 3 previous conversations with a business, **When** they start a new voice call, **Then** the Memory Retrieval Agent returns the most recent conversation summary and the 2 semantically closest past interactions within 500ms of call start.
2. **Given** a customer requests data deletion via the API, **When** the deletion job runs, **Then** all four memory layers return zero records for that customer within 72 hours and a deletion certificate is generated.
3. **Given** a Starter-tier tenant's conversation count reaches 8,000 (80% of 10K limit), **When** the threshold is crossed, **Then** an email alert is sent to the tenant admin within 1 hour.
4. **Given** two channels (voice and WhatsApp) write simultaneously to the same customer's working memory, **When** the Redis WATCH/MULTI transaction executes, **Then** the write succeeds for one channel and the second channel reads the updated state; no data is lost or overwritten incorrectly.

AC4: Sub-Tenant Architecture

1. **Given** Partner P1 creates Sub-Tenant C1, **When** C1 makes an API call, **Then** the RLS policy returns only rows where `tenant_id = C1.id`; no rows from any other tenant are included.
2. **Given** Partner P1 queries their analytics dashboard, **When** the query runs, **Then** data from all of P1's sub-tenants is visible but no data from P2's sub-tenants appears.
3. **Given** Sub-Tenant C1 incurs 2,000 voice minutes of usage in a month, **When** billing runs, **Then** C1's usage appears in P1's invoice as a line item and P1 is charged at the agreed rate; C1's admin sees the same 2,000-minute figure in their own dashboard.
4. **Given** an automated tenant isolation test runs post-deploy, **When** the test queries Tenant A's data as Tenant B, **Then** the test asserts zero rows returned and the deploy proceeds; if rows are returned, the deploy is blocked and an alert is sent.

AC5: Email Intelligence Engine

1. **Given** an inbound email is received on the tenant's connected inbox, **When** it arrives, **Then** an auto-acknowledgment with a ticket ID is sent to the sender within 60 seconds.
2. **Given** an email contains the In-Reply-To header referencing an existing thread, **When** processed, **Then** the email is correctly appended to the existing conversation record, not created as a new case.
3. **Given** an email case has been in the Growth tier queue for 6 hours (75% of 8-hour SLA), **When** the SLA timer triggers, **Then** a yellow alert badge appears in the supervisor dashboard and the assigned agent receives an in-app and email notification.
4. **Given** an outbound email campaign is sent, **When** a recipient clicks the unsubscribe link, **Then** the customer is added to the suppression list within 5 seconds and receives a confirmation message; no further emails are sent.

AC6: Webhook System — Delivery Guarantees

1. **Given** a `booking.confirmed` event occurs, **When** the webhook processor runs, **Then** the HTTP POST to the tenant's configured endpoint is attempted within 1 second of the event.
2. **Given** the tenant's webhook endpoint returns a 500 error, **When** the delivery fails, **Then** the system retries with exponential backoff: 2s, 4s, 8s, 16s, 30s; after 5 failures, the webhook is marked failed and the tenant is notified via email.
3. **Given** a webhook payload is delivered, **When** the tenant's system computes `HMAC-SHA256(raw_body, tenant_secret)`, **Then** it must match `X-AgentCore-Signature`; if timestamp is >5 minutes old, the payload must be rejected.

AC7: Rate Limiting

1. **Given** a Starter-tier tenant sends 150 API requests within 1 minute (limit: 100/min), **When** the 101st request arrives, **Then** the API returns HTTP 429 with `Retry-After: 23` (seconds until token bucket refills) within 50ms.
2. **Given** a tenant has reached their concurrent call limit (e.g., 5 for Starter), **When** a 6th call attempt arrives, **Then** the caller hears a "high volume" message and is offered a callback; no existing calls are disrupted.

AC8: Billing & Metering

1. **Given** a tenant has used 4,000 of their 5,000 included minutes (80% threshold), **When** the 4,000th minute is logged, **Then** an email alert is sent to the tenant admin within 1 hour.
2. **Given** the billing period ends on the last day of the month, **When** the invoice generation job runs at midnight, **Then** a PDF invoice with itemized voice minutes, WhatsApp messages, and API calls is generated and emailed to the tenant's billing contact within 2 hours.
3. **Given** a tenant's payment fails on Razorpay, **When** the failed payment webhook is received, **Then** the tenant receives an email notification, services continue for 7 days grace period, and a second charge attempt is made at D+3.

AC9: Compliance Engine — DPDP and TRAI

1. **Given** TRAI mode is enabled for a tenant, **When** any voice call starts, **Then** the disclosure message "This call is from [Business Name] and may be recorded for quality purposes" plays within the first 5 seconds before the AI proceeds.
2. **Given** a customer's number is on the NDNC registry, **When** the Proactive Outbound Engine attempts to dial, **Then** the call is blocked, the skip is logged with timestamp and customer ID, and no dial attempt is made.
3. **Given** HIPAA mode is enabled for a healthcare tenant, **When** any conversation log or analytics report is generated, **Then** no Protected Health Information (patient name, diagnosis, date of birth) appears in any log accessible to non-clinical staff.

AC10: Agent Copilot

1. **Given** an agent receives a new customer chat message, **When** the message is processed, **Then** the Copilot panel shows 3 ranked suggested replies within 1,000ms.
2. **Given** the customer's message is semantically related to 2 knowledge base articles, **When** the semantic search runs, **Then** the top 2 articles appear in the Copilot KB panel with relevance score and excerpt within 1,000ms.
3. **Given** an agent types a draft reply containing the phrase "I can't help with that," **When** the tone checker detects a negative phrase, **Then** a warning banner appears with a suggested rewrite ("I'd be happy to connect you with someone who can assist with this") before the Send button is enabled.

PART 9: NON-FUNCTIONAL REQUIREMENTS

9.1 Performance

Metric	Target	Measurement Method
Voice end-to-end latency (cascaded)	<800ms p95	Per-call telemetry; OpenTelemetry trace
Voice end-to-end latency (S2S)	<400ms p95	Per-call telemetry
Chat response time (AI)	<2,000ms p99	Chat message timestamp → first token sent
API response time	<200ms p99	ALB access logs
Widget load time	<300ms on 4G	Lighthouse synthetic monitoring
Memory retrieval (all 4 layers)	<500ms p99	Dedicated memory span in trace
Webhook delivery	<1,000ms p99	Webhook delivery timestamp

9.2 Reliability

Metric	Starter	Growth	Scale	Enterprise
Monthly uptime SLA	99.5%	99.9%	99.95%	99.99%
RTO (Recovery Time Objective)	60 min	15 min	5 min	5 min
RPO (Recovery Point Objective)	1 hour	15 min	1 min	1 min
Multi-AZ deployment	No	Yes	Yes	Yes

- Active-passive failover for all stateful services (RDS, Redis)
- Health checks every 30 seconds via ALB
- Automatic ECS task restart on crash (max 3 restart attempts in 5 minutes before alerting)
- CloudWatch alarms for all critical services; PagerDuty on-call rotation for P1/P2 alerts
- Monthly chaos engineering exercise: simulate AZ failure, RDS failover, Redis failover

9.3 Security

- **Encryption at rest:** AES-256 for all PostgreSQL data (RDS encryption enabled), all S3 objects, all Redis AOF logs
- **Encryption in transit:** TLS 1.3 minimum for all API endpoints, WebSocket connections, and inter-service communication; TLS 1.2 deprecated
- **RBAC:** Four default roles: Super Admin, Tenant Admin, Supervisor, Agent. Custom roles configurable. Permissions at resource + action level (e.g., conversations:read, invoices:write). Least-privilege by default.
- **API key security:** Keys are hashed (bcrypt) before storage; never exposed in logs; rotatable without service interruption; IP whitelist option per key
- **Audit log:** Immutable write-only log of all admin actions, API calls, and compliance events; stored in separate RDS instance; cannot be modified or deleted by tenant; 7-year retention
- **Penetration testing:** Annual third-party VAPT; quarterly internal automated scanning (OWASP ZAP, Trivy for container vulnerabilities)
- **Certifications roadmap:** HIPAA BAA — Month 6; SOC 2 Type I — Month 9; SOC 2 Type II — Month 18; ISO 27001 — Month 24; PCI-DSS SAQ-A — Month 12

9.4 Scalability

Dimension	Current Capacity	Scale Target
Concurrent voice calls	200 (1× c6i.xlarge)	1,000 (5× c6i.xlarge auto-scaling)
Concurrent chat sessions	1,000 per ECS task	10,000 (10 tasks)
Active tenants	1,000	10,000+
pgvector embeddings per tenant	1M	10M+ (HNSW index)
API requests/second (platform)	1,000	10,000 (horizontal ECS scaling)
S3 recordings/documents	Unlimited (S3 native)	Unlimited
Webhook events/second	500	5,000 (SQS-backed queue)

Scaling triggers:

- ECS voice workers: scale out when `concurrent_calls_per_task > 50`; scale in at `<20` with 5-minute cooldown
- ECS API workers: scale out when CPU `>70%` for 3 minutes; scale in at `<30%` for 5 minutes
- RDS: scale storage automatically; read replica for analytics queries

9.5 Accessibility

- WCAG 2.1 AA compliance for all web UI components
- All form inputs have visible `<label>` elements (no placeholder-only labels)
- All non-text content (icons, images) have descriptive `alt` or `aria-label` attributes
- Full keyboard navigation: every interactive element reachable and operable via keyboard

- Focus indicators: visible :focus-visible ring on all interactive elements (minimum 2px, 3:1 contrast ratio with adjacent color)
- Screen reader compatibility: tested monthly with NVDA (Windows) and VoiceOver (macOS/iOS)
- Color is never the sole indicator of status: status badges include both color and text/icon
- No flashing content: animations below 3Hz or provide motion-reduce CSS preference support

PART 10: PLATFORM & DEPLOYMENT REQUIREMENTS

10.1 Three Deployment Modes

Model A — Platform API:

- Customer receives: REST API + WebSocket API + OpenAPI docs + SDK
- Customer builds: their own UI and workflow integration
- AgentCore provides: AI voice/chat/WhatsApp intelligence, memory, engines
- Pricing: INR 10,000–1,00,000/month based on usage tier
- Ideal for: Fintechs, healthcare platforms, enterprise IT teams building AI features into existing products
- SLA: 99.99% API uptime for Enterprise; 99.95% for Scale
- Integration support: dedicated technical account manager for Enterprise; self-serve documentation for Starter/Growth

Model B — White-Label:

- Customer receives: Full AgentCore platform deployed on their subdomain (aiagent.clientcompany.com)
- Customer sees: Their branding (logo, colors, domain); no AgentCore branding visible to their end users
- AgentCore provides: Full platform including agent desktop, supervisor dashboard, admin portal, all 24 engines
- Pricing: INR 25,000/month base + INR 1,000–2,000/sub-tenant/month
- Ideal for: BPOs, system integrators, insurance companies, large enterprises
- Sub-tenant support: full hierarchical tenancy; white-label partner manages their own client base
- Partner portal: dedicated portal for partner to manage sub-tenants, view aggregated analytics, generate reports

Model C — Vertical SaaS Embedded:

- AgentCore embedded as the AI layer within HelpRevX vertical products
- ItinerarySuite: full lead → booking → invoice → NPS lifecycle with AgentCore voice AI
- PhotoAdda: studio booking, package quoting, follow-up AI

- aayiq.com: Intercom-style engagement with AgentCore as the AI resolution layer
- aayiq.ai: standalone PLG brand (see Part 11)
- Pricing: bundled with vertical product; AgentCore costs baked into vertical SaaS pricing
- Customer experience: seamless AI without knowing it's "AgentCore"

10.2 Independent Deployment Spec

Can AgentCore run completely standalone, independent of any HelpRevX product? YES.
AgentCore is a fully self-contained platform.

Minimum System Requirements for Independent Deployment:

- **Application servers:** 2× AWS EC2 c6i.xlarge (4 vCPU, 8GB RAM) in separate AZs; or equivalent on GCP/Azure
- **Database:** PostgreSQL 15+ with pgvector extension; RDS db.t3.large minimum (2 vCPU, 8GB); Multi-AZ recommended
- **Cache:** Redis 7.0+; ElastiCache cache.t3.medium minimum
- **Storage:** S3 (or S3-compatible: MinIO for self-hosted); minimum 100GB initial
- **LiveKit Server:** 1× EC2 c6i.xlarge reserved (\$49/mo)
- **Network:** VPC with public subnets (ALB) and private subnets (application/database layers); NAT Gateway

Docker Compose for Local Development:

- `docker-compose.yml` includes: PostgreSQL with pgvector, Redis, LocalStack (S3 emulation), LiveKit Server, Mailhog (email testing), AgentCore API service, AgentCore Worker service
- Single command: `docker-compose up` → working local environment within 5 minutes
- Pre-loaded with seed data: 1 demo tenant, 5 knowledge articles, 1 configured voice persona

Terraform Modules for AWS Production:

- `modules/vpc`: VPC, subnets, routing, NAT Gateway, security groups
- `modules/ecs`: ECS cluster, task definitions, service definitions, auto-scaling policies, ECR repositories
- `modules/rds`: PostgreSQL RDS instance, read replica, parameter groups, subnet groups
- `modules/redis`: ElastiCache Redis cluster, replication group, subnet groups
- `modules/livekit`: EC2 reservation, security groups, EBS volumes
- `modules/cdn`: CloudFront distribution, S3 origins, SSL certificate (ACM)
- `modules/monitoring`: CloudWatch dashboards, alarms, SNS topics, PagerDuty integration

10.3 Multi-Tenancy Requirements

Data isolation guarantees:

- PostgreSQL Row Level Security (RLS) enforced at database level; bypass only for platform admin role
- Every database connection sets `app.current_tenant_id` via `SET LOCAL` at transaction start
- Application code never queries without tenant context; enforced by ORM-level middleware
- Automated isolation test runs on every deployment; blocks deploy if isolation test fails
- Separate S3 prefixes per tenant for all file storage (`s3://agentcore-prod/{tenant_id}/...`)
- Redis keys namespaced per tenant (`{tenant_id}:session:{session_id}`)

Tenant provisioning API:

```
POST /api/v1/tenants
{
  "name": "ACME Insurance",
  "plan": "scale",
  "parent_tenant_id": "uuid|null",
  "admin_email": "admin@acme.com",
  "subdomain": "acme",
  "industry": "bfsi"
}
→ 201 Created
{
  "tenant_id": "uuid",
  "api_key": "ak_live_...",
  "subdomain": "acme.agentcore.ai",
  "status": "pending",
  "onboarding_url": "https://acme.agentcore.ai/onboarding"
}
```

PART 11: aayiq.ai — NEW STANDALONE PRODUCT RECOMMENDATION

11.1 Product Concept

aayiq.ai is the consumer-friendly, PLG-first brand for AgentCore's SMB market. Where AgentCore is sold to mid-market and enterprise buyers through a consultative sales process, aayiq.ai is designed for the solo entrepreneur, photography studio owner, or coaching center director who needs AI for their business today and wants to sign up with a credit card in 15 minutes.

The product concept: "The AI Agent for Indian Businesses." Think of it as Intercom Fin for India — a product that non-technical business owners can understand, configure, and use without an IT team. Every aayiq.ai product capability runs on AgentCore infrastructure under the hood.

11.2 Positioning

	AgentCore	aayiq.ai	Intercom Fin	aayiq.com
Target buyer	CTO, CX Head, Ops Manager	Business owner, solo entrepreneur	Product/CX teams	Marketing/Sales teams
Sales motion	Enterprise B2B	PLG self-serve	Mid-market sales	SMB/mid-market
Setup time	Days–weeks	<30 minutes	Days	Hours
Price entry point	₹10,000/month	₹999/month	\$29/seat/month	₹499/month
Indic language	First-class	First-class	None	Partial
Voice AI	Yes	Yes	Limited	No

[aayiq.ai](#) differentiates from Intercom Fin on price (4–8× cheaper) and Indic language capability. It differentiates from [aayiq.com](#) by being AI-first (autonomous agent resolution rather than human-assisted engagement). It is the "entry drug" into the AgentCore ecosystem.

11.3 Target Customer

SMBs across all 15 AgentCore industries with these characteristics:

- Revenue: ₹10 lakh to ₹10 crore/year
- Team size: 1–25 people
- Technology comfort: uses WhatsApp, Google Suite, basic CRM
- Decision timeline: same day to 1 week (card-on-file signup)
- Primary need: "I miss calls. I forget to follow up. I need help at night."
- Discovery channel: Google search ("AI for business India"), YouTube, Instagram ads, referral from peer businesses

11.4 Pricing

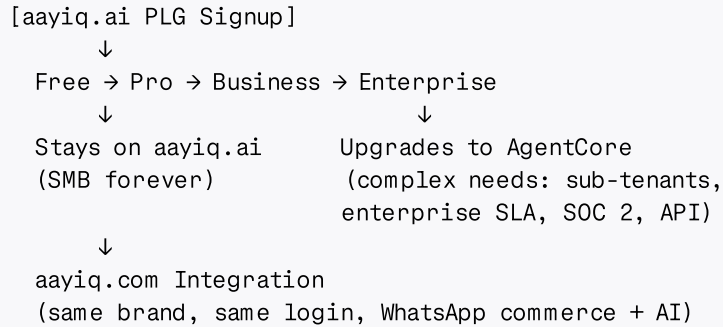
Tier	Monthly Price	Included Minutes	WhatsApp Messages	Use Case
Free	₹0	100 voice minutes	200 messages	Try it; personal projects
Pro	₹999/month	1,000 voice minutes	2,000 messages	Freelancers, solo businesses
Business	₹4,999/month	10,000 voice minutes	20,000 messages	Growing SMBs, 5–20 person teams
Enterprise	Custom	Custom	Custom	Fast-growing SMBs outgrowing Business tier; bridge to AgentCore

Overage pricing:

- Voice: ₹4.50/minute (above included)
- WhatsApp: ₹0.80/message (above included)

No per-seat pricing. [aayiq.ai](#) charges for usage, not headcount. This removes the primary friction point in SMB SaaS adoption.

11.5 How [aayiq.ai](#) Relates to the Ecosystem



Technical relationship: [aayiq.ai](#) is a Model C deployment of AgentCore. The [aayiq.ai](#) frontend (separate React app, [aayiq.ai](#) domain) calls AgentCore APIs under the hood. Every [aayiq.ai](#) tenant is an AgentCore tenant tagged with source: `aayiq.ai`. Upgrades to AgentCore Enterprise require only a plan change — all data, configurations, and conversation history migrate automatically.

Brand relationship: [aayiq](#) (Hindi: "come") is the parent brand. [aayiq.com](#) handles engagement/marketing. [aayiq.ai](#) handles intelligence/automation. Both share brand identity, auth (single login), and the customer's data. The combined pitch: "aayiq helps Indian businesses communicate better — [aayiq.com](#) for your marketing, [aayiq.ai](#) for your AI agent."

11.6 Launch Plan for [aayiq.ai](#)

Pre-Launch (Month 1–2):

- Build [aayiq.ai](#) marketing site with clear value proposition and pricing
- Implement simplified onboarding wizard (3 steps, not 5): Industry → Connect WhatsApp → Test message
- Set up self-serve billing (Razorpay subscription + credit card)
- Create 5 industry-specific demo videos (travel, photography, healthcare, education, real estate)
- Set up referral program (refer a business, both get 500 free minutes)

Beta Launch (Month 3):

- Invite 100 beta users from existing HelpRevX network
- Target: 50 active users with ≥50 AI conversations each by end of Month 3
- Weekly feedback sessions; rapid iteration cycle (1-week sprints)

Public Launch (Month 4):

- Google Search Ads targeting "AI for business India," "WhatsApp chatbot India," "voice bot for small business"
- YouTube pre-roll ads (Hindi and English)
- Instagram/Facebook ads targeting business owners in Tier 1 and Tier 2 cities
- Press release to YourStory, Inc42, Economic Times Tech

Month 6 targets: 1,000 paying customers, ₹50 lakh MRR from [aayiq.ai](#) alone.

PART 12: OPEN QUESTIONS

#	Question	Owner	Blocking?	Target Resolution
Q1	Which BSP do we use for WhatsApp at launch — AiSensy or direct Meta Cloud API? AiSensy offers onboarding support; direct API offers lower per-message cost at scale.	CTO	YES — blocks WhatsApp Engine build	May 2026
Q2	Will we build our own Agent Copilot suggested-reply model or use GPT-4.1-mini streaming? Own model = lower cost at scale; GPT = faster to market.	Product + ML	NO — GPT as default, revisit at 10M conversations/month	June 2026
Q3	What is the SLA for NDNC registry API response time? If NDNC check adds >500ms latency to outbound calls, we may need to batch-check and cache results.	Backend Engineer	NO	May 2026
Q4	Does HIPAA BAA require us to self-host all AI models (no sending PHI to OpenAI)? Legal interpretation needed for DPDP healthcare context.	Legal + CTO	YES — blocks healthcare go-live	May 2026
Q5	Should aayiq.ai have its own Postgres schema or separate RDS instance? Separate instance = isolation and upsell path; same schema = simpler operations.	Backend Architect	NO	May 2026
Q6	What is the partner (white-label) margin structure? Do partners pay wholesale and charge retail? Or do they receive a revenue share of end-client ARR?	CEO + Sales	YES — blocks white-label sales	April 2026
Q7	Can we support Facebook Messenger as a channel in Phase 1? Meta's approval process and same API as WhatsApp suggests yes; but UI/UX scope increases.	Product	NO — defer to Phase 2	June 2026
Q8	Should the Itinerary Builder Engine support B2C sharing (customer-facing itinerary page) or only B2B (agent portal only)?	Product	NO	May 2026
Q9	What is the data retention policy for voice recordings for healthcare tenants? HIPAA requires 6-year medical record retention, but storage costs increase.	Legal + Finance	YES — blocks HIPAA BAA	May 2026

#	Question	Owner	Blocking?	Target Resolution
Q10	Should we support Salesforce integration natively in Phase 1, or only via webhook/Zapier? Salesforce native = large enterprise credibility; webhook = faster to ship.	Product + CTO	NO	June 2026
Q11	Is Exotel's API rate limit sufficient for an enterprise tenant running 200 concurrent calls? Need confirmation of Exotel Enterprise tier limits.	Infrastructure	YES — blocks Enterprise voice SLA	April 2026
Q12	How do we handle cases where a customer's WhatsApp number and voice number are different? Customer identity merge logic needs specification.	Product	NO	May 2026
Q13	What is our approach to AI hallucination guard-rails in the Proposal and Quote Engines? Incorrect pricing in a generated quote creates legal liability.	Product + ML	YES — blocks Proposal/Quote Engine launch	May 2026
Q14	Should we offer Telegram as a channel? Telegram Bot API is well-documented; relevant for NGO and legal verticals.	Product	NO — Phase 3	Q3 2026
Q15	What is the minimum viable Partner Portal for Model B launch? Full sub-tenant management, or just white-label branding with billing?	Product	YES — blocks Model B	May 2026
Q16	Do we need a dedicated data processing environment for EU customers to comply with GDPR data residency requirements?	Legal + Infrastructure	NO — defer to first EU customer	Q3 2026
Q17	How do we enforce minimum margins in the Quote Engine without exposing supplier cost data to all agents? RBAC on cost fields or masked display?	Product	NO	June 2026
Q18	Should aayig.ai support English only at launch to reduce scope, with Hindi added in Month 2?	Product	NO — launch English + Hindi simultaneously	April 2026
Q19	What is the process for tenants to submit and manage WhatsApp template approvals? Do we mediate with Meta via AiSensy or provide a direct portal?	Product	YES — blocks WhatsApp commerce	May 2026
Q20	Should the NPS survey be sent immediately after call end, or delayed 2 hours to allow customer sentiment to settle? Research suggests delayed surveys have higher completion rates.	Product	NO	June 2026

PART 13: APPENDIX

13.1 Glossary

Term	Definition
AgentCore	HelpRevX's universal agentic AI platform. The technical infrastructure behind all AI interactions.
aayiq.ai	Consumer brand for AgentCore; PLG-first, SMB-targeted AI agent product.
AHT	Average Handle Time — average duration of a customer interaction (voice, chat, or email)
Barge-In	Ability for the AI to stop speaking immediately when the customer starts talking
BSP	Business Solution Provider — Meta-certified partner for WhatsApp Business API (e.g., AiSensy)
Cascaded Pipeline	Voice architecture: STT → LLM → TTS sequential processing; 600–1000ms latency
Circuit Breaker	Pattern that stops calling a failing service after N failures; prevents cascade failures
CSAT	Customer Satisfaction Score — typically 1–5 star rating after an interaction
CES	Customer Effort Score — measures ease of interaction; lower effort = higher score
DPDP	Digital Personal Data Protection Act 2023 — India's primary data privacy law
Engine	A discrete, orchestratable AI capability unit within AgentCore (e.g., Booking Engine, Lead Engine)
FCR	First Contact Resolution — percentage of issues resolved in the first interaction without callback
HMAC	Hash-based Message Authentication Code — used for webhook signature verification
HNSW	Hierarchical Navigable Small World — index algorithm for fast approximate nearest-neighbor search in pgvector
IVFFlat	Inverted File with Flat quantization — pgvector index for smaller vector datasets
LangGraph	Framework for building multi-agent AI orchestration graphs (Supervisor pattern)
LiveKit	Open-source WebRTC infrastructure for real-time voice and video
NPS	Net Promoter Score — measures customer likelihood to recommend (Promoters – Detractors)
NDNC	National Do Not Call — TRAI's registry of numbers that have opted out of telemarketing
Noisy Neighbor	Problem where one tenant's resource usage degrades performance for other tenants
pgvector	PostgreSQL extension for vector similarity search (semantic memory)
PLG	Product-Led Growth — go-to-market strategy where the product itself drives user acquisition
RLS	Row Level Security — PostgreSQL feature for tenant data isolation at the database level
RPO	Recovery Point Objective — maximum acceptable data loss; measured in time
RTO	Recovery Time Objective — maximum acceptable downtime; measured in time
S2S	Speech-to-Speech — direct audio processing via OpenAI Realtime API; 200–400ms latency
Sarvam-M	Sarvam AI's 105B parameter multilingual model for 11 Indic languages
Shadow DOM	Web standard for encapsulating component styles and preventing CSS conflicts

Term	Definition
STT	Speech-to-Text — transcribing spoken audio to text
Tenant	A business or organization using AgentCore as their AI platform
TRAI	Telecom Regulatory Authority of India — regulates telemarketing calls and SMS
TTS	Text-to-Speech — converting text to spoken audio
VAD	Voice Activity Detection — determines when a person has started or stopped speaking
VAPT	Vulnerability Assessment and Penetration Testing
WebRTC	Web Real-Time Communication — browser-standard protocol for voice/video P2P connections
White-Label	A product sold by one company but branded and resold by another

13.2 Competitive Feature Matrix

Feature	AgentCore	Intercom Fin	Genesys CX	Freshdesk	AiSensy	Bland AI
AI Voice Agent	✓ Hybrid pipeline	✗	✓ Limited	✗	✗	✓ English only
Hindi/Indic Language	✓ 11 languages	✗	✗	✗	Partial	✗
WhatsApp Commerce	✓ Native Flows + UPI	✗	✗	✗	✓	✗
4-Layer Memory	✓ Redis+pg+pgvector	✗ Conversation only	✗	✗	✗	✗
Multi-Agent Orchestration	✓ LangGraph 7 workers	✗	Limited	✗	✗	✗
24 CRM Lifecycle Engines	✓	✗ 5 basic	✓ Workflow	Partial	✗	✗
Sub-Tenant Architecture	✓	✗	✓	✗	✗	✗
Agent Copilot	✓	✓	✓	Partial	✗	✗
DPDP / TRAI Compliance	✓ Native	✗	✗ India	✗ India	Partial	✗
HIPAA Support	✓ Month 6	✓	✓	✓	✗	✗
Self-Hosted Deployment	✓	✗	✗	✗	✗	✗
White-Label	✓	✗	✓	✗	✗	✗
India Entry Price	₹10,000/mo	~₹12,000/mo	~₹1,25,000/mo	₹2,000/agent	₹2,000/mo	\$500/mo
Proactive Outbound AI	✓	Partial	✓	✗	✗	Partial
Document Intelligence	✓	✗	✗	✗	✗	✗

Feature	AgentCore	Intercom Fin	Genesys CX	Freshdesk	AiSensy	Bland AI
NPS / CSAT Automation	✓	✗	✓	✓	✗	✗
Itinerary / Vertical Engines	✓ Travel	✗	✗	✗	✗	✗
SDKs (iOS/Android/RN)	✓	✓	✓	Partial	✗	✗
Webhook + HMAC Sig	✓	✓	✓	✓	✗	✗

13.3 Decision Log

#	Decision	Rationale	Date	Alternatives Considered
D1	Use LiveKit Agents (Python) instead of Vonage/Twilio Voice SDK	LiveKit is open-source and self-hostable, reducing per-minute costs from ~\$0.01/min (Twilio) to infra cost only. Python SDK aligns with LangGraph/ML stack.	Feb 2026	Twilio Voice SDK, Vonage Voice API, Agora
D2	Use Sarvam-M 105B as primary Indic LLM (free tier) rather than GPT-4.1 for Indian language conversations	Cost savings of ~\$0.40/M tokens at scale; Sarvam-M outperforms GPT-4.1-mini on Indic language benchmarks per Sarvam's published evaluations	Feb 2026	GPT-4.1 for all, Google Gemini 2.0, Claude Haiku
D3	Implement hybrid cascaded + S2S pipeline with mid-call switching	Cascaded enables per-component tuning and lower cost; S2S enables lowest latency when needed. No competitor offers dynamic switching.	Feb 2026	S2S-only (higher cost), cascaded-only (higher latency)
D4	Use PostgreSQL with pgvector for semantic memory rather than a dedicated vector DB (Pinecone, Weaviate)	Eliminates a separate database, reduces operational complexity, and pgvector with HNSW performs comparably to dedicated vector DBs below 100M vectors	Mar 2026	Pinecone (costly at scale), Weaviate (additional ops), Qdrant
D5	Three-level tenant hierarchy (Platform → Partner → End Client) capped at 3 levels	3 levels covers all white-label use cases. Deeper hierarchies create exponential RLS complexity and performance degradation.	Mar 2026	2 levels (too restrictive for white-label), unlimited levels (RLS unmanageable)
D6	Use ECS EC2 (not Fargate) for voice agent workers	Voice is stateful; Fargate's ephemeral containers cause context loss on task replacement. ECS EC2 allows sticky task placement and persistent memory for active calls.	Feb 2026	Fargate (considered and rejected), Kubernetes EKS (higher ops burden)
D7	Use AiSensy BSP for WhatsApp (vs. direct Meta Cloud API)	AiSensy provides faster business verification onboarding, managed WABA (WhatsApp Business Account), and India-specific support. At current scale, AiSensy per-message fees are marginal.	Mar 2026	Direct Meta Cloud API (considered for Year 2 when volume justifies), 360dialog

#	Decision	Rationale	Date	Alternatives Considered
D8	Launch aayiq.ai as a standalone product, not a feature of aayiq.com	Standalone brand enables dedicated PLG motion, separate pricing page, and independent product identity. aayiq.com is B2B engagement; aayiq.ai is B2B AI resolution — different buyers.	Apr 2026	Extension of aayiq.com (considered and rejected — brand confusion, pricing complexity)
D9	Shadow DOM isolation for chat widget	Prevents host-page CSS from breaking widget UI without requiring iframe (which introduces sandboxing limitations for WebRTC and file upload). CSP-compatible.	Mar 2026	iframe (CSP conflicts, WebRTC limitations), no isolation (too fragile)
D10	Alembic for database migrations (not Prisma, not Flyway)	Python-native (matches API/worker stack); supports complex migration sequences; widely used in FastAPI/SQLAlchemy ecosystem; zero-downtime migration patterns well-documented	Mar 2026	Prisma Migrate (Node.js only), Flyway (JVM), Liquibase

End of Document

Document Control:

Version	Date	Author	Changes
0.1	March 2026	Product Team	Initial draft
0.8	March 30, 2026	Product + Audit	Incorporated V3 audit findings
1.0	April 2026	Product Team	Final — all 15 critical gaps resolved, 33 engines specified, aayiq.ai recommendation added

Next Review: July 2026 (Q2 retrospective + Phase 2 planning)

Distribution: Engineering All-Hands, Product Team, Design Team, Sales (executive summary only), Investor Data Room

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